PFC Examples

The PFC Examples Application: setting it up, and what each example shows.

The PFC Examples Application 3
Setting up the Examples Application 4
The main window 5
The application's files 6
Adding an example 7
Application Services 8
  Date / Time Service 9
  Error Service Example 11
  File Services Example 12
  Language Service 13
  Logical Unit of Work 14
  Metaclass Service 15
  Security Service 16
  Timing Service 17
  Transaction Registration 18
  Utility Services 19
DataWindow Services 20
  DataStore Services 22
  DDDW Calculator and Calendar 23
  Drop-down Search Service 24
  Dynamic Filter Expressions 25
  Filter Service Dialogs 26
  Filter Service Settings 27
  Find and Replace Service 28
  Grid Service 29
  Multiple Table Update 30
  Print Preview Service 31

Query Mode Service 32
Refresh Drop-down DataWindows 33
Report Service 34
Required Columns Service 35
Resize Freeform DataWindow 36
Resize Grid DataWindow 37
Row Manager Service 38
Row Selection Service 39
Sort Service 40
Linkage Services 41
Across Tab Pages 42
Basic Filters 43
Basic Retrieval Arguments 44
Basic Scrolling 45
Filter with Cascading Deletes 46
Filter with Cascading Keys 47
Retrieval with Auto-updates 48
TreeView and Listview 49
Basic Listview 50
Basic Treeview 51
Recursive Tree View 52
Treeview Linked to a Listview 53
Update a Treeview 54
Window Services 55
Resize Dynamic Tab Pages 56
Ribbon Bar 57
Save Process: All DW Scenarios 58
Sheet Manager 59
Status Bar 60
Weighted Resize of a Window 61
Window Resize Service 62
Objects 63

Controls Gallery 64
Find and Replace in a RTE 65
Progress Bar Sampler 66
Split Bar 3 Pane Style 67
Split Bar Explorer Style 68
Split Bar Settings 69
Type Ahead in the DDLB & DDPLB 70

The PFC Examples Application

The PFC Examples Application shows the PowerBuilder Foundation Classes at work. Each example is a small window that uses one PFC service or object: open it, try it, then open the window in the PowerBuilder IDE to see how it's done. The windows are short, and their code can be copied into your own PFC applications.

The main window lists the examples in a tree, by category:

- Application Services: Services that the application manager starts, and the utility objects that any object can use.
- DataWindow Services: The services of PFC's DataWindow control, `u_dw`, and of its DataStore, `n_ds`. Each is started with an `of_Set` function, such as `of_SetSort`(TRUE).
- Linkage Services: The linkage service (`n_cst_dwsrv_linkage`) links DataWindows in master-detail relationships: by retrieving the details with retrieval arguments, by filtering them, or by scrolling to them.
- TreeView and Listview: PFC's TreeView and ListView controls with services (`u_tvs` and `u_lvs`) fill themselves from DataWindow objects, DataStores or SQL, one level at a time.
- Window Services: The services of PFC windows: the save process, resizing, the status bar, the sheet manager, and a frame with a ribbon bar.
- Objects: PFC's visual user objects: split bars, the progress bar, type-ahead list boxes, the Rich Text Edit control, and a gallery of the newer controls.

Select an example and press F1, or press F1 in an example's window, to read about that example.

In this help

- Setting up the Examples Application: the database, the INI file and the images
- The main window
- The application's files
- Adding an example

The PFC documentation, the User's Guide and the Object Reference, is in the docs repository.

Setting up the Examples Application

The project is `pfc examples`: `pfc examples.pbproj` in the PFC 2025 Solution, or the `pfc examples` target in the PFC 2025 Workspace. It builds `pfcexamp.exe`.

The database

Most examples read and update the EASDEMO sample database, from the Database-scripts-and-files repository:

- **SQL Server:** restore `EASDEMO.sql_server.bak` as the database EASDEMO.
- **SQL Anywhere:** use `easdemo17.db` with SQL Anywhere 17, through an ODBC data source. PowerBuilder's own demo database (the PB Demo DB data source) has the same tables and works too.

Point the `[Database]` section of `examples.ini`, in the `examples` folder, at it: `DBMS`, `DbParm`, and for SQL Server `ServerName`, `LogId` and `LogPassword`. Keep `AutoCommit=False`. The Security Service example also needs a `[Security]` section for the PFC security database (`PFC.sql_server.bak` or `pfc17.db`); it says so when it opens without one.

The images

The examples' pictures are in the PFC-Example-App-Image-Files repository. Add its folder to the system path or to PowerBuilder's application path, so that both the IDE and `pfcexamp.exe` find them.

This help file

Put `PFCEXAMP.chm` beside `pfcexamp.exe`. It is published, with a PDF version, in the Example-application-help-file repository.

The main window

The main window, `w_examplemain`, is itself a PFC example. It uses:

- the TreeView level source service (`u_tvs`, `n_cst_tvsrv_levelsource`) for the tree of categories and examples, and the ListView data source service (`u_lvs`, `n_cst_lvsrv_datasource`) for the list beside it;
- the example catalog, `n_cst_examplecatalog`, which reads the tree and the list from `exam.txt` and `examcat.txt` (see Adding an example);
- the debug and SQL Spy services, from the Utilities menu;
- the most recently used (MRU) service;
- the window resize service, with a split bar between the tree and the list.

Double-click an example, or select it and press Enter, to open it. Each example window takes its title from the catalog. Press F1 to read about the selected example.

The application's files

Libraries

The examples are in the `examples` folder:

Library	Contents
`appexamp.pbl`	The application object (`pfcexamp`) and the example windows
`appexmdw.pbl`	The examples' DataWindow objects
`appexmfe.pbl`	The main window, its menu and the example catalog
`exmmain.pbl`, `exmutil.pbl`, `exmwnsrv.pbl`	The application's extension layer: its own `w_master`, `w_frame`, `n_tr` and other objects that inherit from PFC's

The application also uses the PFC libraries (`pfcapsrv`, `pfcdwsrv`, `pfcmain`, `pfcutil`, `pfcwnsrv`) and their extension layers (`pfeapsrv`, `pfedwsrv`, `pfemain`, `pfeutil`, `pfewnsrv`).

Other files

File	Contents
`examples.ini`	The application's INI file: the database connection and the application's settings
`exam.txt`, `examcat.txt`	The example catalog
`README.md`	Setting up, the catalog files and the list of examples

Adding an example

The main window's tree and list come from two tab-delimited files in the `examples` folder, without a header line, with CR LF line ends.

`exam.txt` has a line for each example: `example_id` (a unique number), `title` (the name in the list, and the window's title), `exhelp` (the help context number of its topic in this file, 1000 + example_id; a category's topic is 100 + category_id) and `window` (the window that the example opens).

`examcat.txt` has a line for each item in the tree: `category_id` (a unique number), `category_name`, `category_parent_id` (the category it's in), `example_id` (0 for a category), the picture indexes (5 and 6 for an example, 3 and 4 for a category), the state picture (0) and the overlay picture (1 marks an example as new).

To add an example:

1. Write its window. Most examples inherit from `w_main`, with a short introduction at the top and a Close button that triggers `pfc_close`.
2. Add a line to `exam.txt`, and a line to `examcat.txt` under its category.
3. Add its topic to this help file (in `src/topics.js` in the Example-application-help-file repository), with the same help context number.

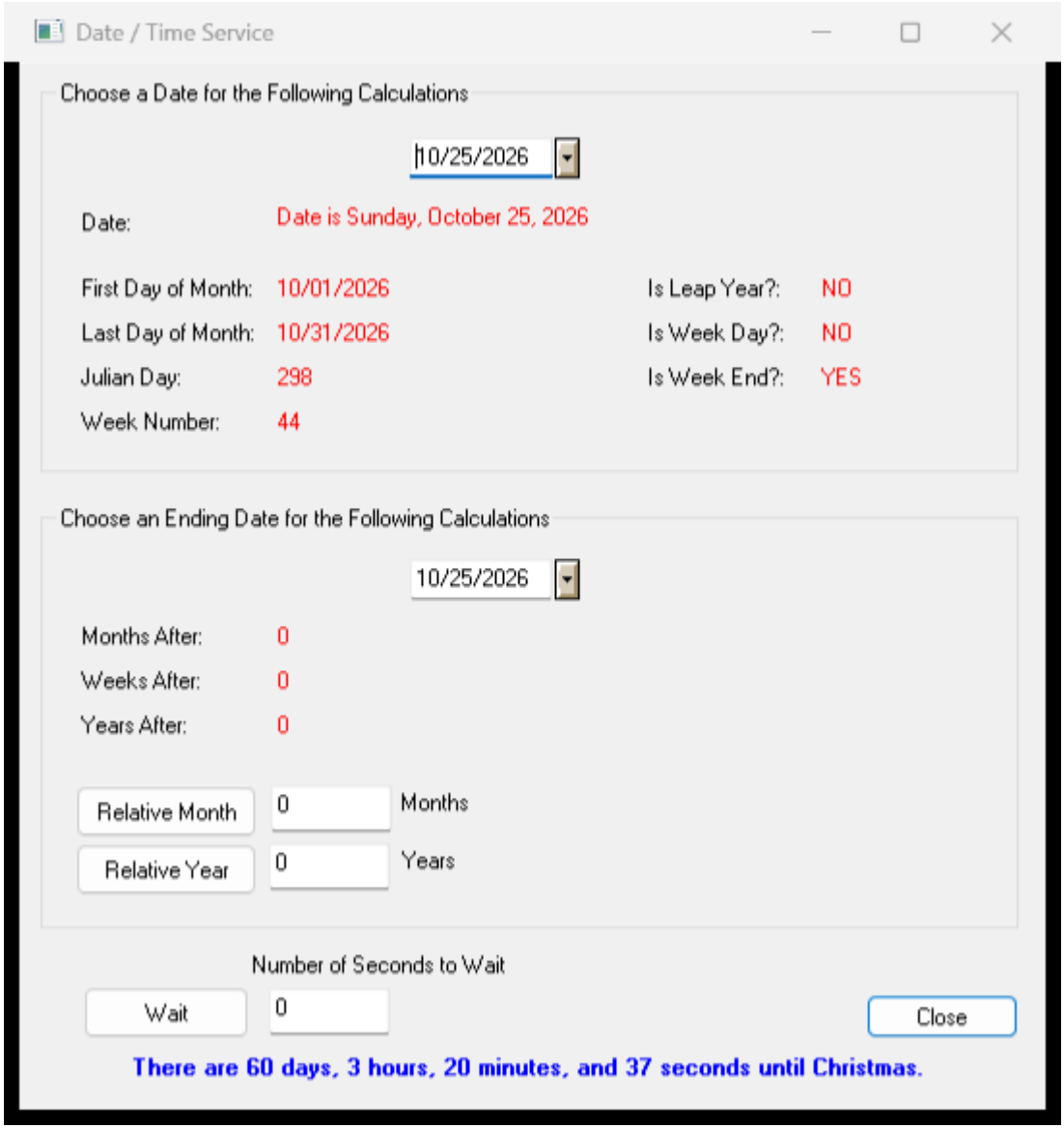
Application Services

Services that the application manager starts, and the utility objects that any object can use.

Example	Window
Date / Time Service	w_dateserv
Error Service Example	w_errormsgservice
File Services Example	w_fileserv
Language Service	w_languageserv
Logical Unit of Work	w_luw
Metaclass Service	w_metaclass
Security Service	w_securityserv
Timing Service	w_timingserv
Transaction Registration	w_trregserv
Utility Services	w_utilityserv

Date / Time Service

The date and time service (`n_cst_datetime`) calculates with dates and times: the first and last day of a month, leap years, weekdays and weekends, Julian day numbers, week numbers, relative months and years, and the months, weeks and years between two dates.



Try it

- Choose a date to see what the service says about it.
- Choose an ending date to see the months, weeks and years after the first date.
- Try Relative Month, Relative Year and Wait.

PFC objects

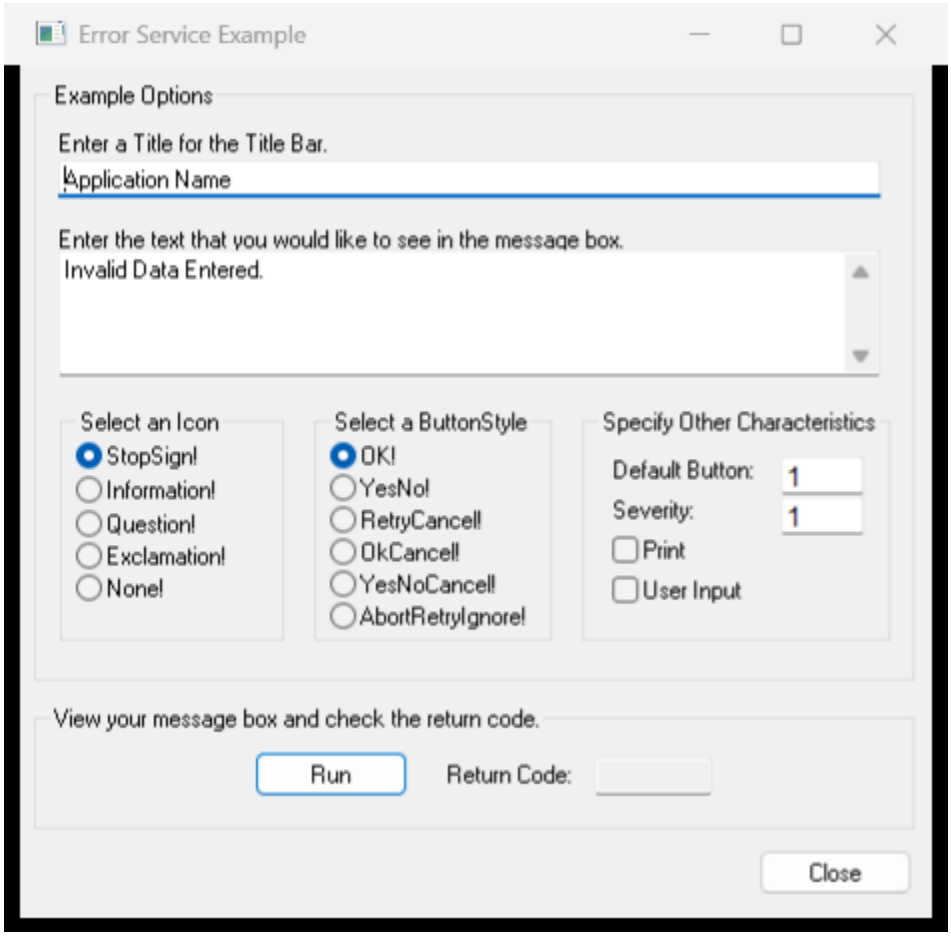
`n_cst_datetime`

The window

`w_dateserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Error Service Example

The error service (`n_cst_error`) displays messages, either with PowerBuilder's MessageBox or in PFC's own message window (`w_message`), which can also print the message and take input from the user.



Try it

- Type a title and a message, choose the icon, the buttons, the default button and the severity, and click Run; the return code shows which button was clicked.
- Choose the PFC style (`of_SetStyle`) and check Print and User Input to see `w_message`'s extra buttons.

PFC objects

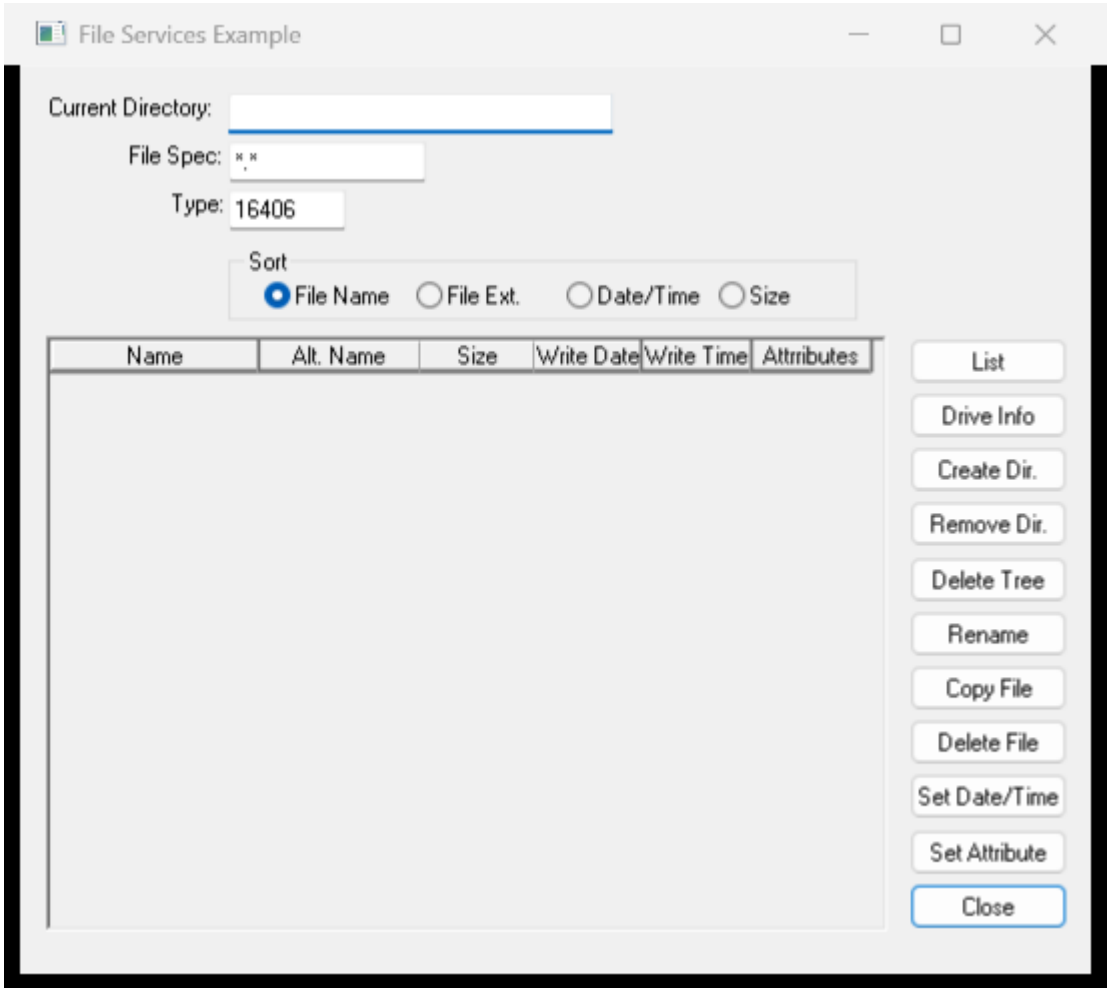
`n_cst_error`, `w_message`

The window

`w_errormsgservice`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

File Services Example

The file service (`n_cst_filesrv`) works with files and directories: it lists a directory, with sizes, dates and attributes, and creates, renames, copies and deletes files and directories.



Try it

- Change the directory and the file specification, and list the files; sort them by name, extension, date or size.
- Use the buttons to create and remove a directory, rename, copy or delete a file, and set its date and attributes. Try them in a folder of your own.

PFC objects

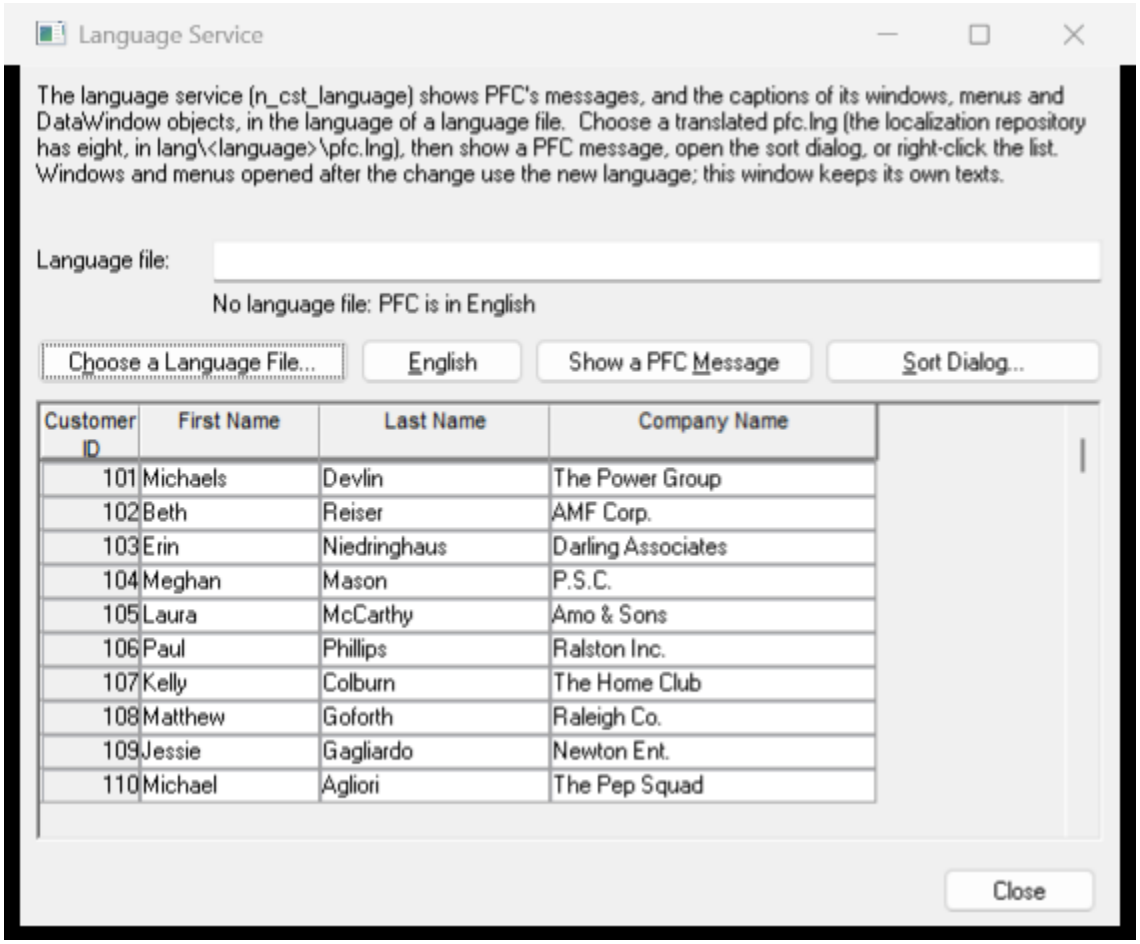
`n_cst_filesrv`

The window

`w_fileserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Language Service

The language service (`n_cst_language`) shows PFC's messages, and the captions of its windows, menus and DataWindow objects, in the language of a language file (pfc.lng).



Try it

- Choose one of the translated language files (the localization repository has eight), then show a PFC message or open a PFC dialog box.

PFC objects

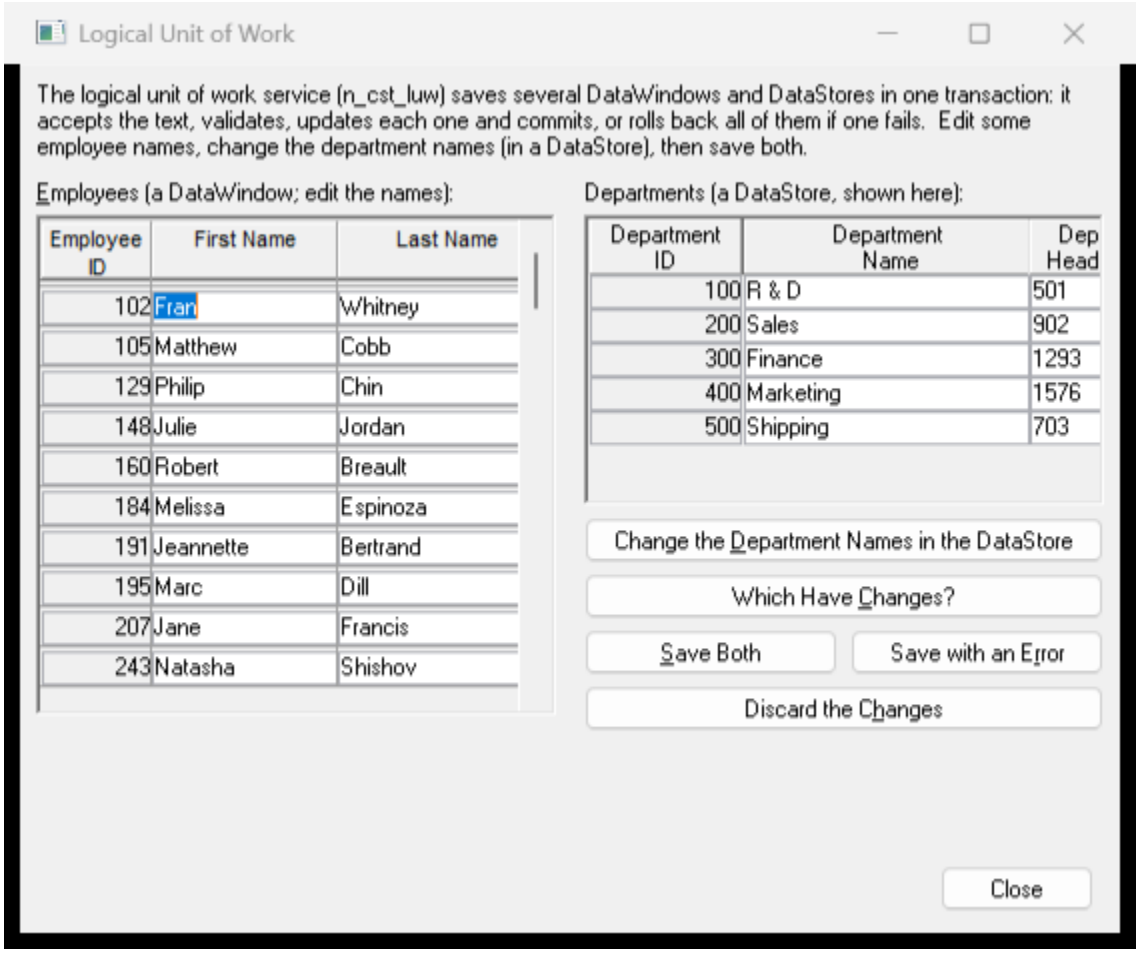
`n_cst_language`

The window

`w_languageserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Logical Unit of Work

The logical unit of work service (`n_cst_luw`) saves several DataWindows and DataStores in one transaction: it accepts the text, validates, updates each one and commits, or rolls back all of them if one fails. Every PFC window's save process (`pfc_save`) uses it.



Try it

- Edit some employee names, and change the department names (they are in a DataStore, shown with `ShareData`).
- Click Which Have Changes? (`of_UpdatesPending`), then Save Both (`of_Save`).
- Click Save with an Error: the departments' update fails after the employees', and the whole transaction is rolled back.

PFC objects

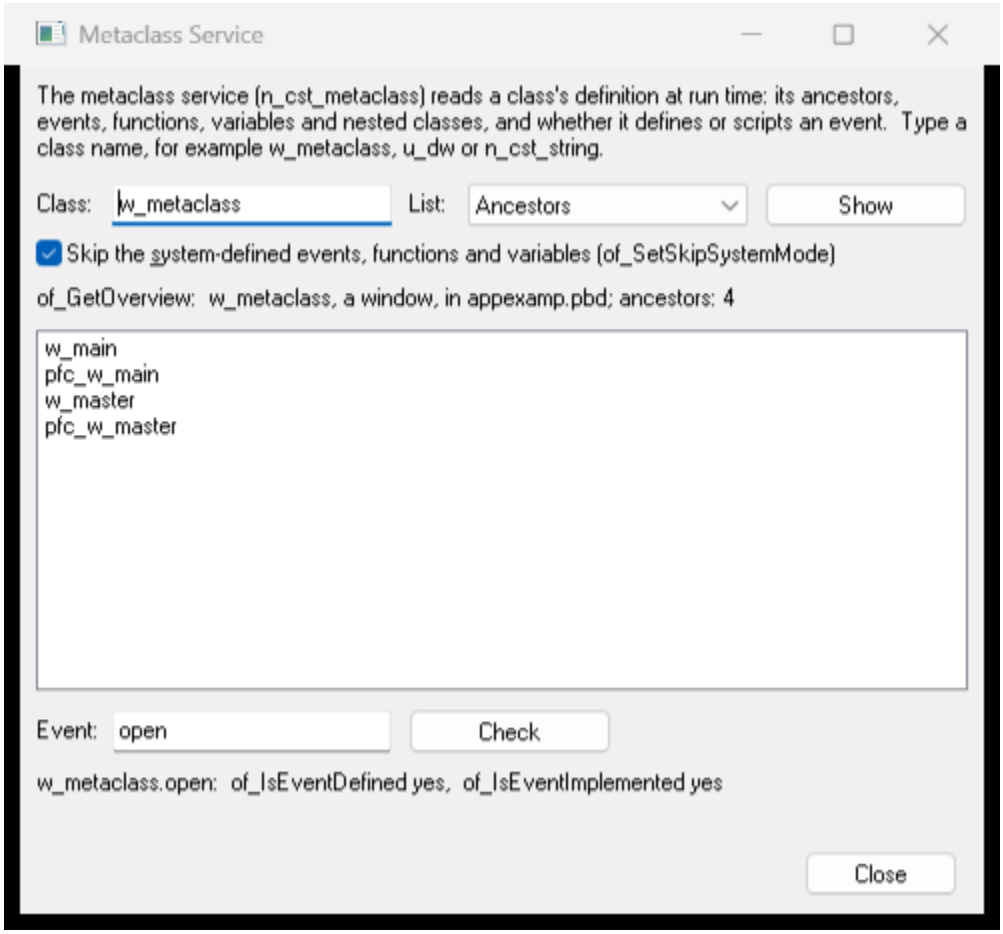
`n_cst_luw`

The window

`w_luw`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Metaclass Service

The metaclass service (`n_cst_metaclass`) reads a class's definition at run time: its ancestors, its events, functions and instance variables, and the classes nested in it, and whether it defines or scripts an event.



Try it

- Type a class name, such as `w_metaclass`, `u_dw` or `n_cst_string`, choose what to list, and click Show.
- Clear Skip the system-defined events, functions and variables to see the ones PowerBuilder defines too (`of_SetSkipSystemMode`).
- Type an event name and click Check to see whether the class defines it (`of_IsEventDefined`) and whether it or an ancestor scripts it (`of_IsEventImplemented`).

PFC objects

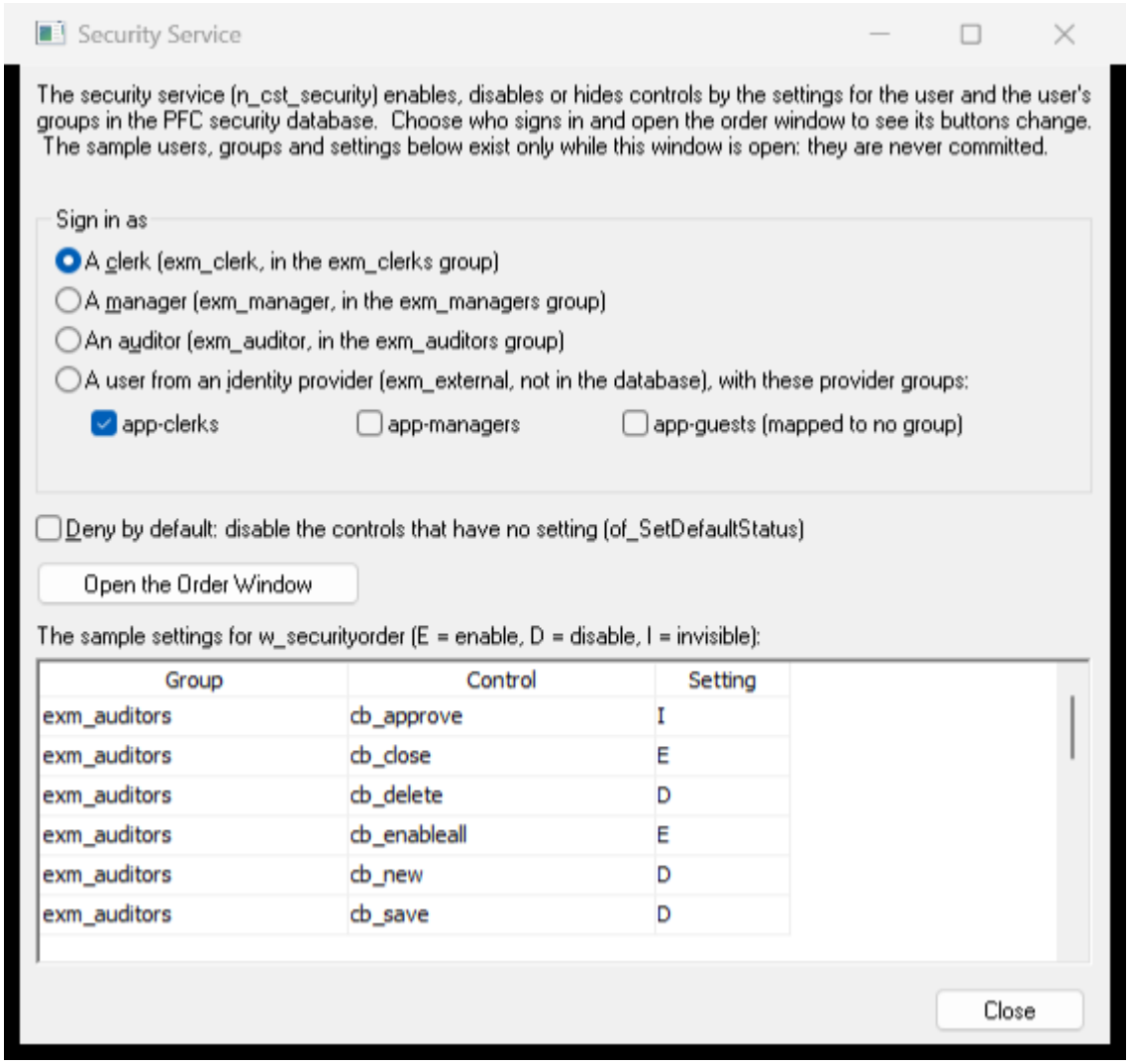
`n_cst_metaclass`

The window

`w_metaclass`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Security Service

The security service (`n_cst_security`) enables, disables or hides a window's controls by the settings for the user, and for the user's groups, in the PFC security database.



Try it

- Choose who signs in, then open the order window to see its buttons change.
- The sample users, groups and settings exist only while the example is open; they are rolled back when it closes.

PFC objects

`n_cst_security`

The window

`w_securityserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Timing Service

PFC's timing object, `n_tmg`, triggers or posts an event in the objects registered with it: in one object, with the single-object service (`n_cst_tmgsingle`), or in several, each at its own interval, with the multiple-object service (`n_cst_tmgmultiple`).



Try it

- Watch the clock (one event, every second) and the three counters (every 2, 4 and 8 seconds, on one timer).
- Stop and start each timer, and unregister the 2-second counter to see the shared timer's interval change.

PFC objects

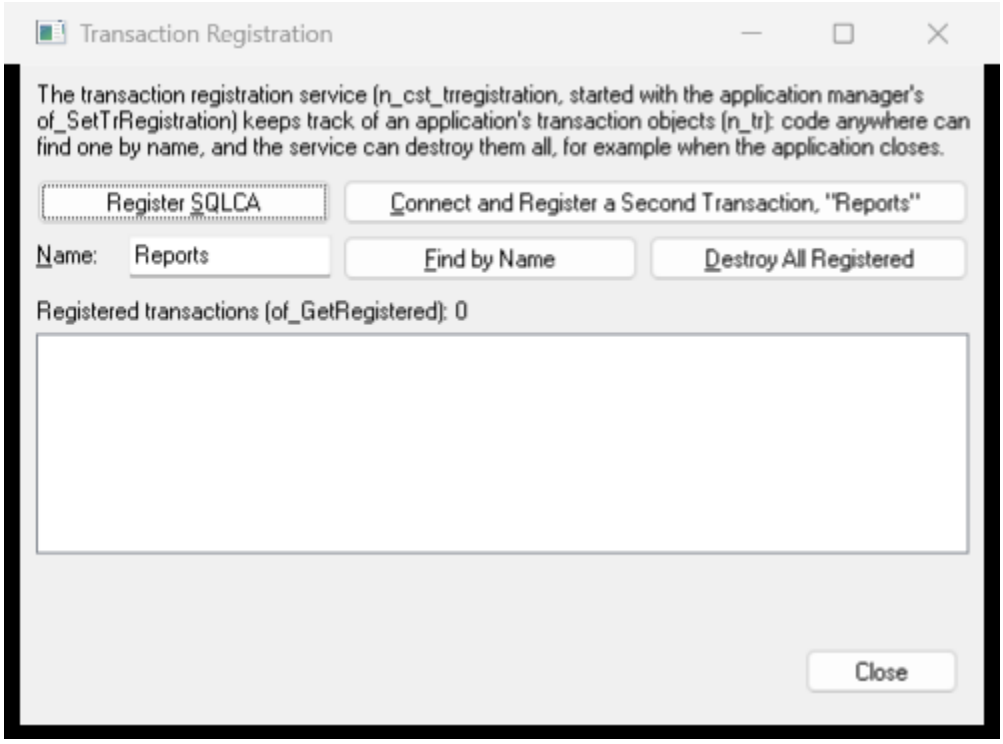
`n_tmg`, `n_cst_tmgsingle`, `n_cst_tmgmultiple`

The window

`w_timingserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Transaction Registration

The transaction registration service (`n_cst_trregistration`, started with the application manager's `of_SetTrRegistration`) keeps track of an application's transaction objects: code anywhere can find one by name, and the service can destroy them all when the application closes.



Try it

- Register `SQLCA`, and connect and register a second transaction named Reports.
- Find a transaction by name (`of_GetByName`), and destroy all the registered ones (`of_Destroy`; `SQLCA` is kept with `of_SetAllowSQLCADestroy`).

PFC objects

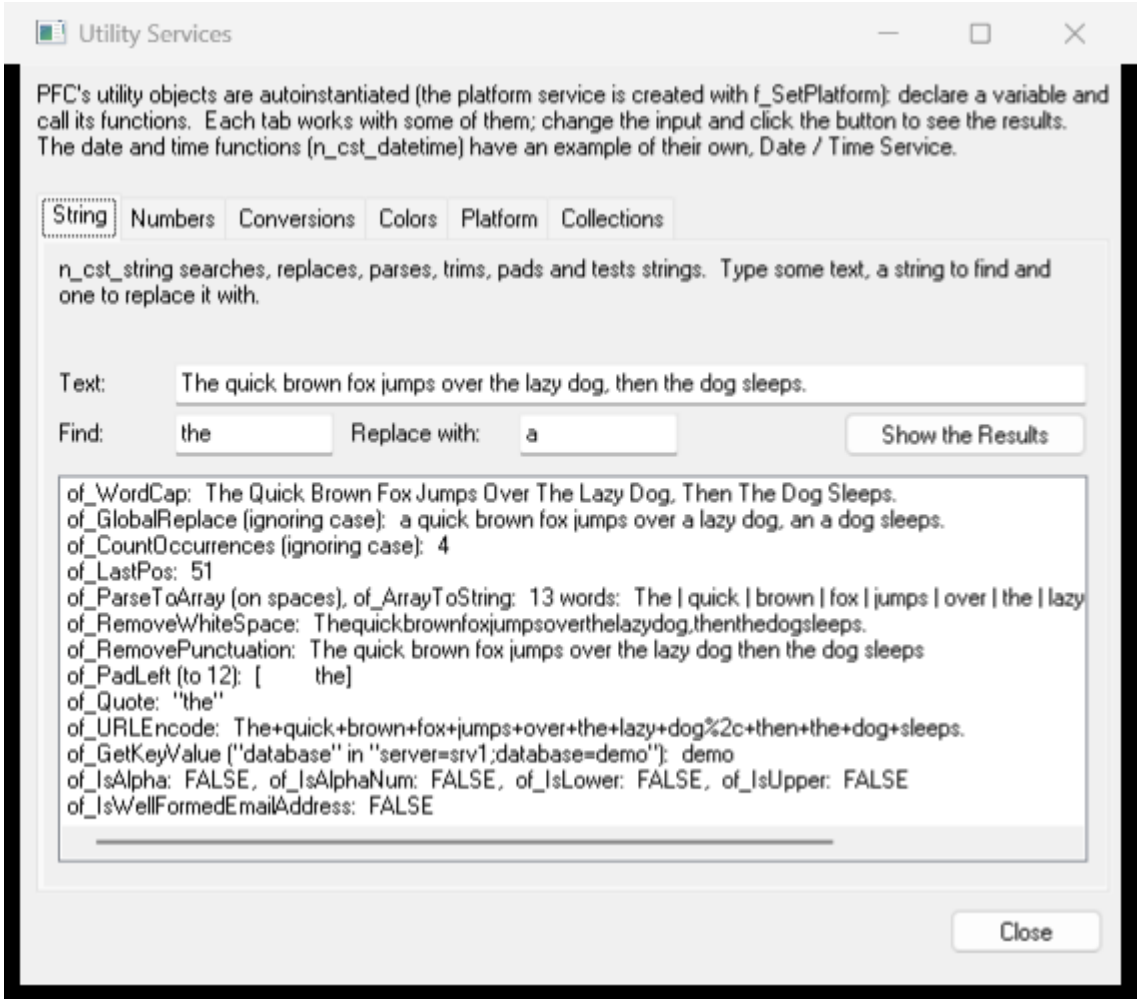
`n_cst_trregistration`, `n_tr`

The window

`w_trregserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Utility Services

PFC's utility objects, with a tab for each group: strings (`n_cst_string`), numbers (`n_cst_numerical`), conversions (`n_cst_conversion`, `n_cst_measureconversion`, `n_cst_pbunitconversion`), colors (`n_cst_color`, `n_cst_colorconv`), the platform (`n_cst_platform`, `n_cst_environment`) and the collections (`n_cst_list`, `n_cst_stack`, `n_cst_queue`, `n_cst_tree`).



Try it

- On each tab, change the input and click the button to see the results of the functions.

PFC objects

`n_cst_string`, `n_cst_numerical`, `n_cst_conversion`, `n_cst_color`, `n_cst_platform`, `n_cst_environment`, `n_cst_list`, `n_cst_tree`

The window

`w_utilityserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

DataWindow Services

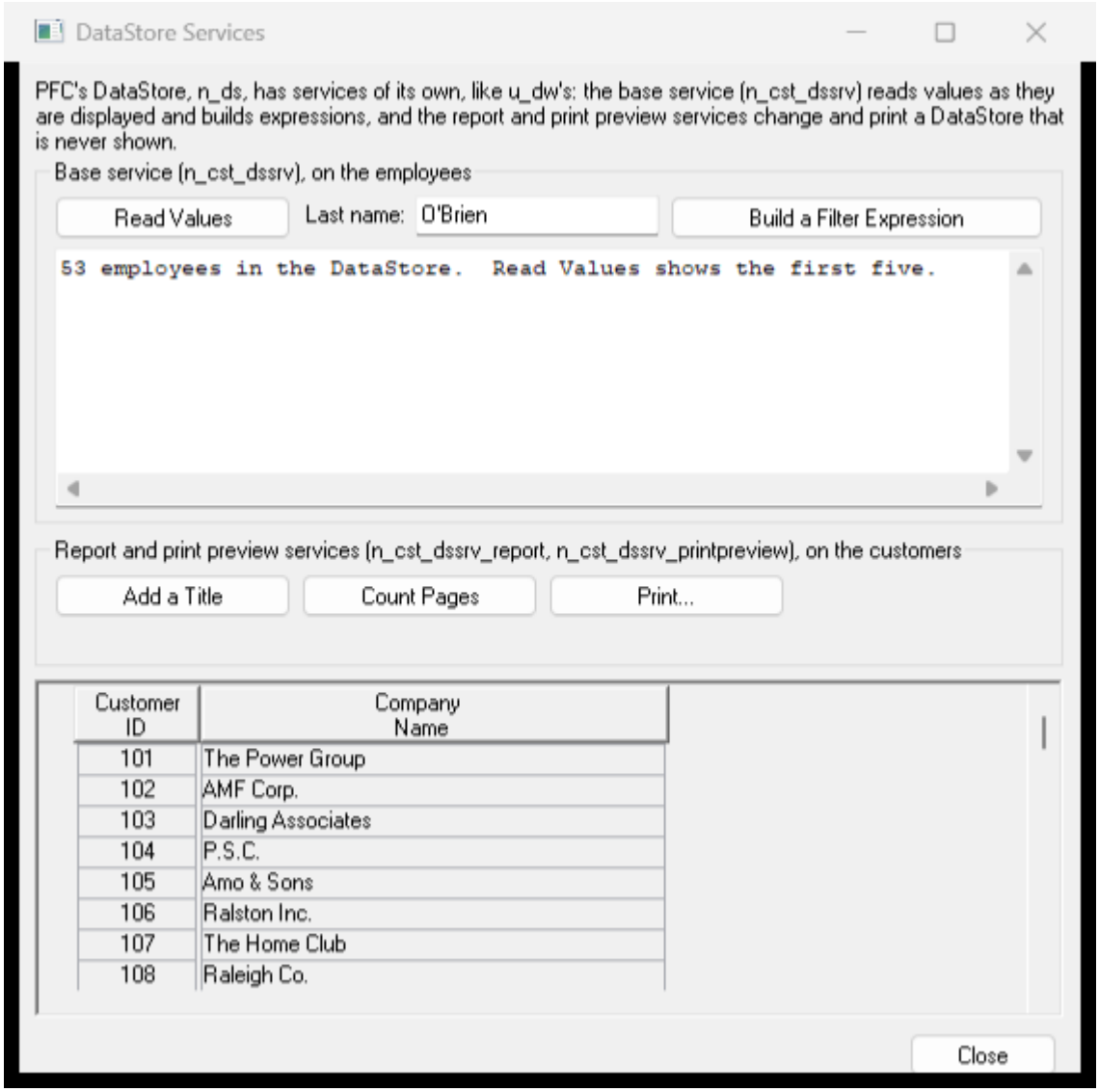
The services of PFC's DataWindow control, `u_dw`, and of its DataStore, `n_ds`. Each is started with an `of_Set` function, such as `of_SetSort`(TRUE).

Example	Window
DataStore Services	`w_dsserv`
DDDW Calculator and Calendar	`w_ddcalsamp`
Drop-down Search Service	`w_dropdownsearchdw`
Dynamic Filter Expressions	`w_buildfilter`
Filter Service Dialogs	`w_filterdialogs`
Filter Service Settings	`w_filtersettings`
Find and Replace Service	`w_findreplacedw`
Grid Service	`w_gridserv`
Multiple Table Update	`w_multiupdate`
Print Preview Service	`w_dwprintpreview`
Query Mode Service	`w_querymode`
Refresh Drop-down DataWindows	`w_refreshdddws`
Report Service	`w_reportserv`
Required Columns Service	`w_requiredfields`
Resize Freeform DataWindow	`w_dwresizeff`
Resize Grid DataWindow	`w_dwresizegrid`
Row Manager Service	`w_rowmanager`

Row Selection Service	`w_rowselection`
Sort Service	`w_sort`

DataStore Services

PFC's DataStore, `n_ds`, has services of its own, like `u_dw`'s: the base service (`n_cst_dssrv`) reads values as they are displayed and builds expressions, and the report and print preview services (`n_cst_dssrv_report`, `n_cst_dssrv_printpreview`) change and print a DataStore that is never shown.



Try it

- Try each button, and see the DataStore's data in the DataWindow that shares it.

PFC objects

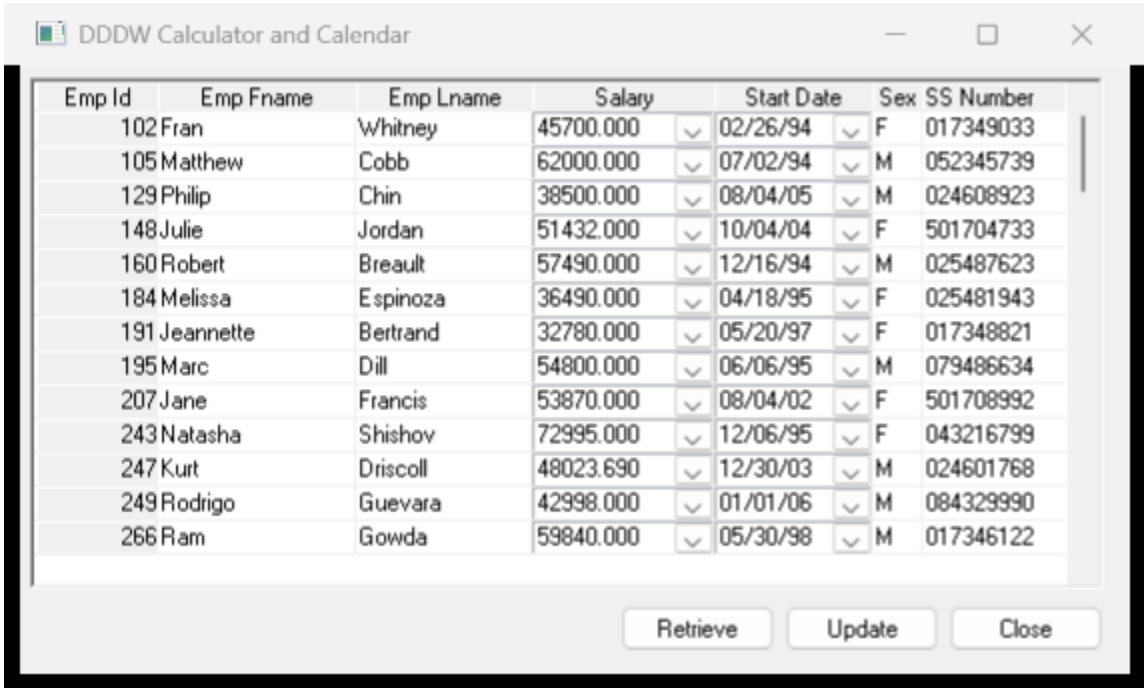
`n_ds`, `n_cst_dssrv`, `n_cst_dssrv_report`, `n_cst_dssrv_printpreview`

The window

`w_dsserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

DDDW Calculator and Calendar

The drop-down calculator and calendar (`u_calculator`, `u_calendar`) for DataWindow columns, started with `of_SetDropDownCalculator` and `of_SetDropDownCalendar`.



Try it

- Click the drop-down arrows of the number and date columns.

PFC objects

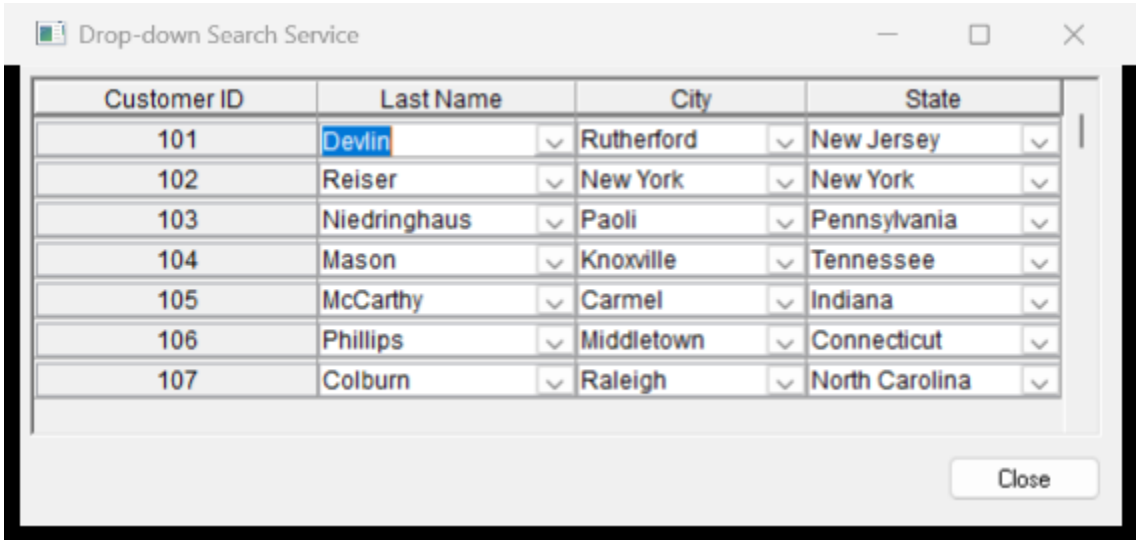
`u_calculator`, `u_calendar`

The window

`w_ddcalsamp`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Drop-down Search Service

The drop-down search service (`n_cst_dwsrv_dropdownsearch`) completes what the user types in a drop-down DataWindow column with the first matching value of its list.



Try it

- Type the first letters of a value in the drop-down columns.

PFC objects

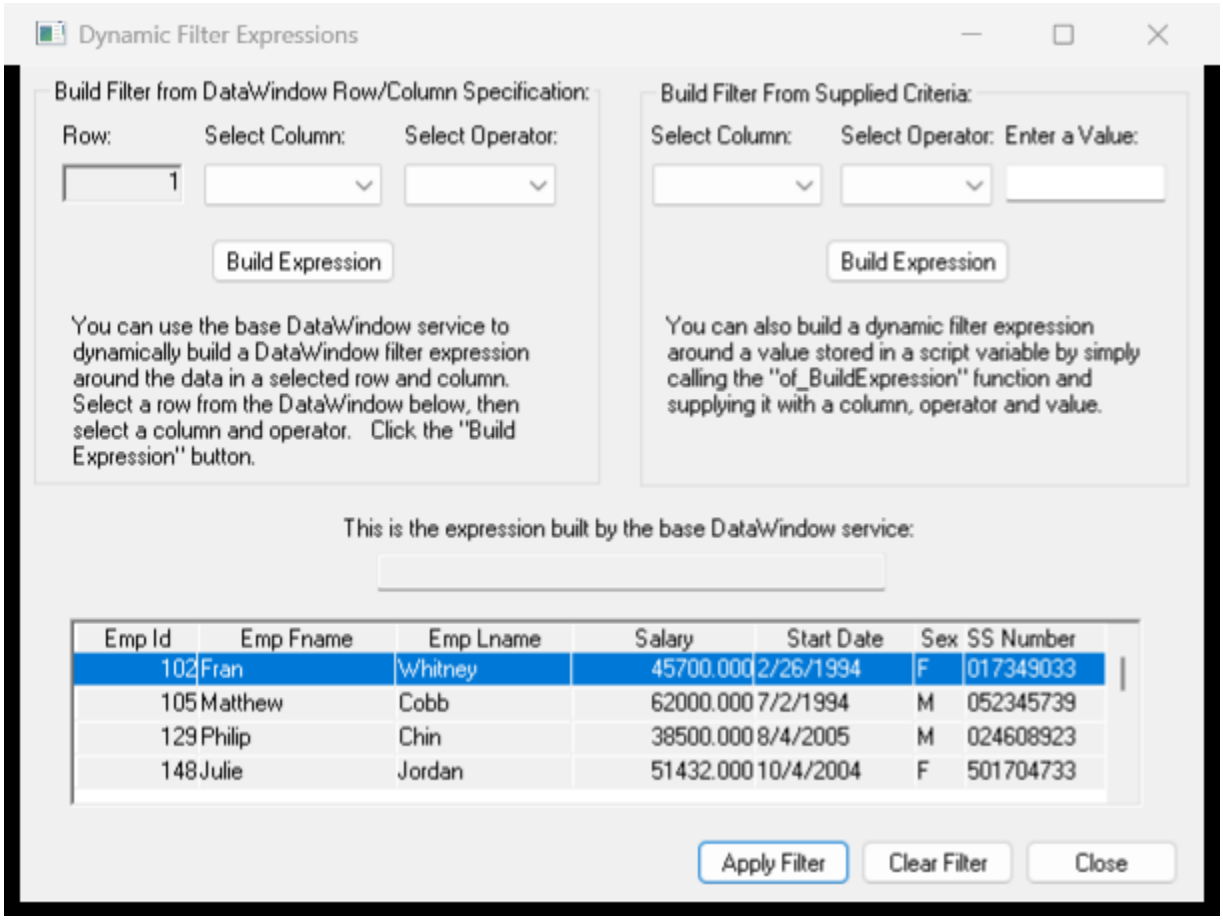
`n_cst_dwsrv_dropdownsearch`

The window

`w_dropdownsearchdw`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Dynamic Filter Expressions

The base DataWindow service builds filter expressions around the data (`of_BuildExpression`): from a row and a column the user selects, or from a value.



Try it

- Select a row, a column and an operator, and build the filter.
- Type a value and build an expression around it.

PFC objects

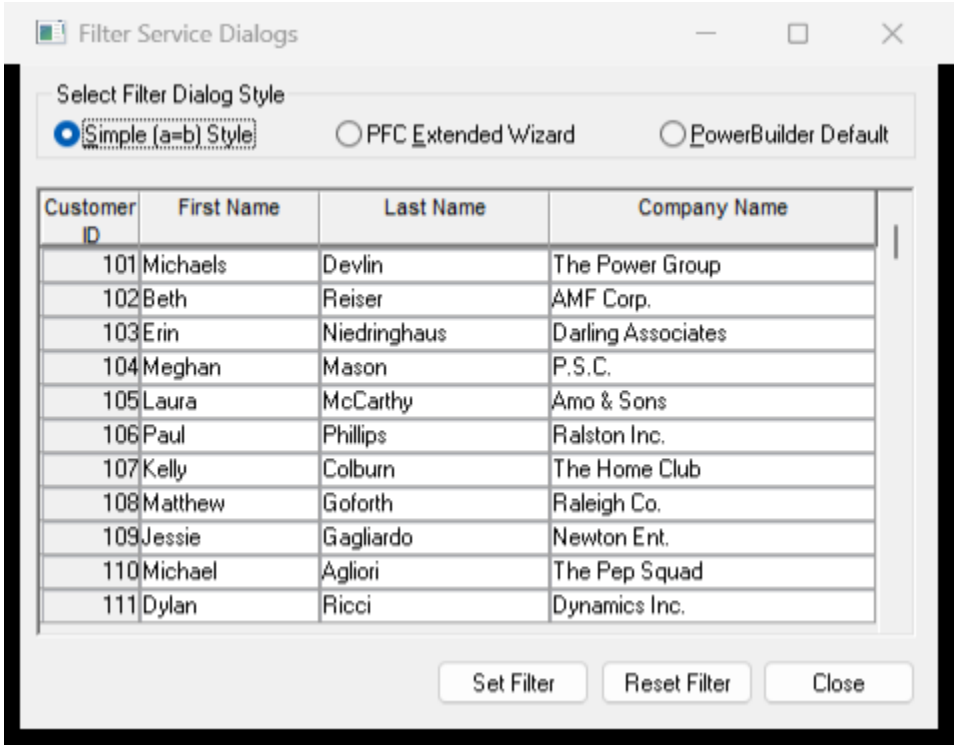
`n_cst_dwsrv`

The window

`w_buildfilter`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Filter Service Dialogs

The filter service (`n_cst_dwsrv_filter`) shows one of three filter dialog boxes: PowerBuilder's, PFC's simple one (column, operator and value) or PFC's extended one, which builds an expression from the functions, columns, operators and values.



Try it

- Choose a dialog style, click Set Filter, and filter the DataWindow; Reset Filter shows all the rows again.

PFC objects

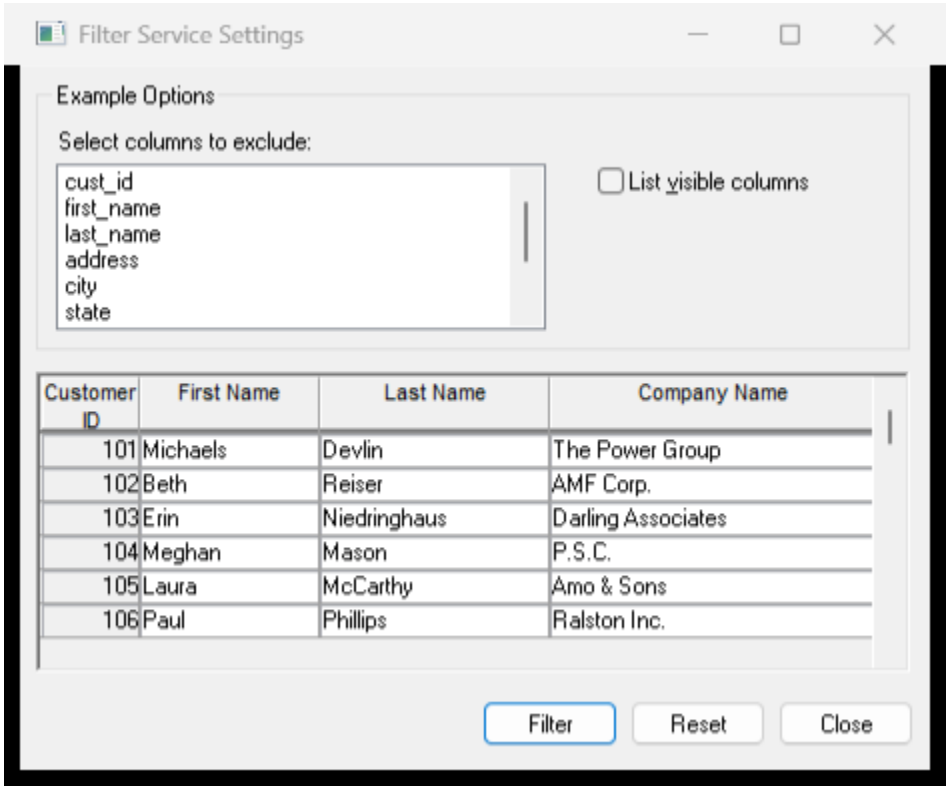
`n_cst_dwsrv_filter`

The window

`w_filterdialogs`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Filter Service Settings

The filter service can show only the visible columns in its dialog boxes, or leave out the columns you choose (`of_SetVisibleOnly`, `of_SetExclude`).



Try it

- Check List Visible Columns, or select columns to exclude, then click Filter.

PFC objects

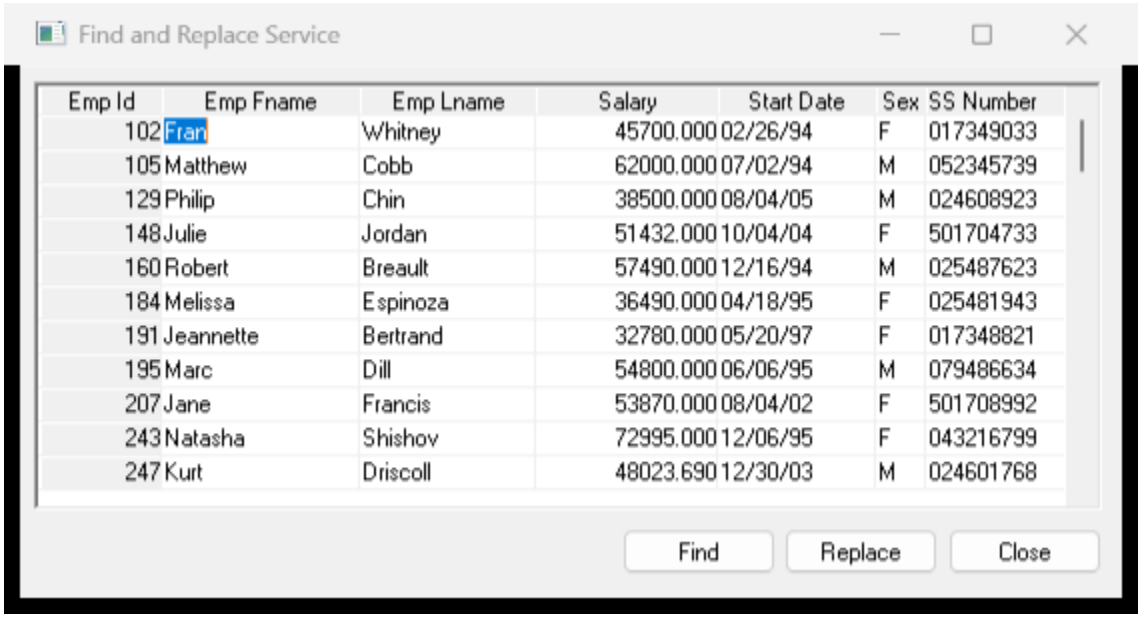
`n_cst_dwsrv_filter`

The window

`w_filtersettings`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Find and Replace Service

The find service (`n_cst_dwsrv_find`) adds Find and Replace dialog boxes to a DataWindow.



Try it

- Click Find, or Replace, and search the DataWindow.

PFC objects

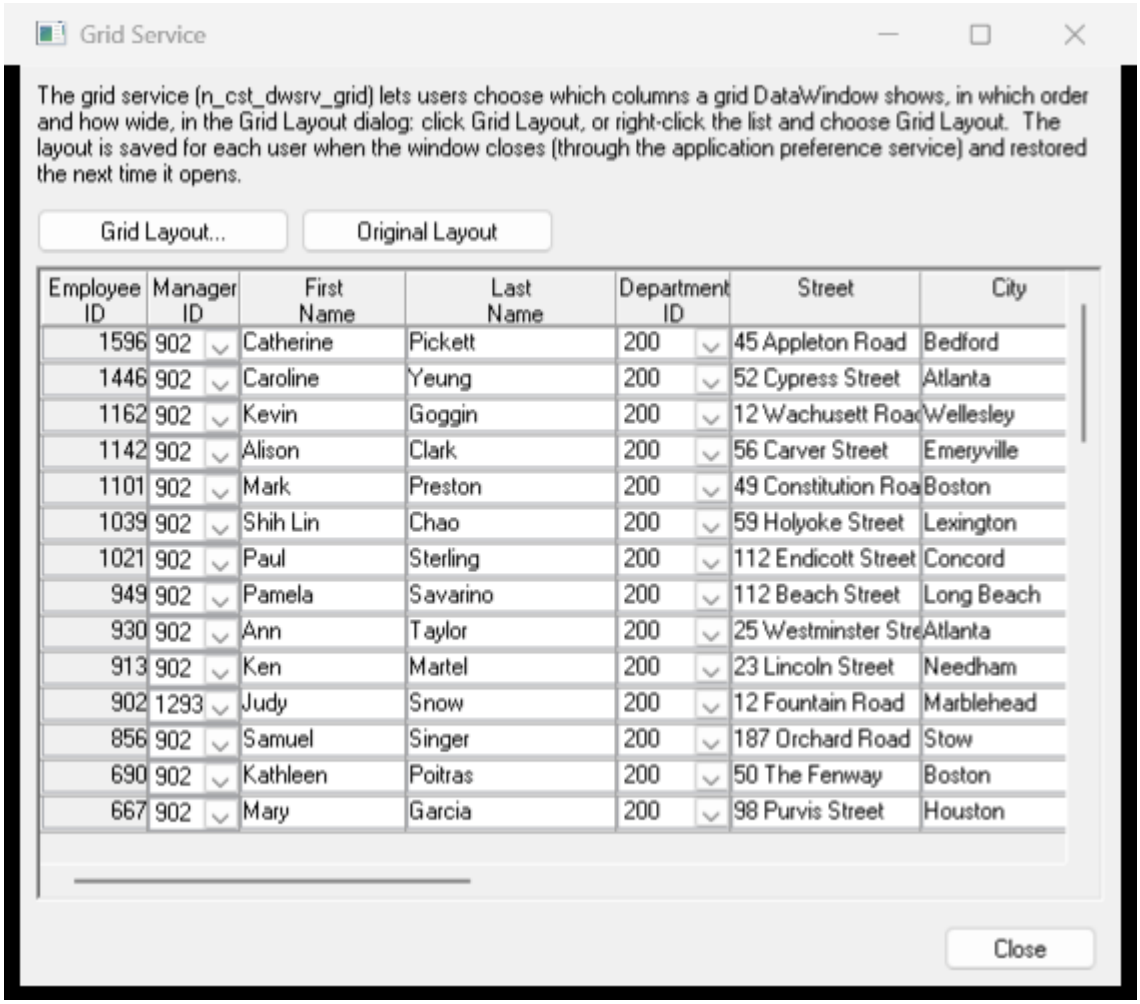
`n_cst_dwsrv_find`

The window

`w_findreplacedw`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Grid Service

The grid service (`n_cst_dwsrv_grid`) lets users choose which columns a grid DataWindow shows, in which order and how wide, in the Grid Layout dialog box. The layout is saved for each user.



Try it

- Click Grid Layout, or right-click the list, hide and move some columns, and close the window: the layout comes back the next time.

PFC objects

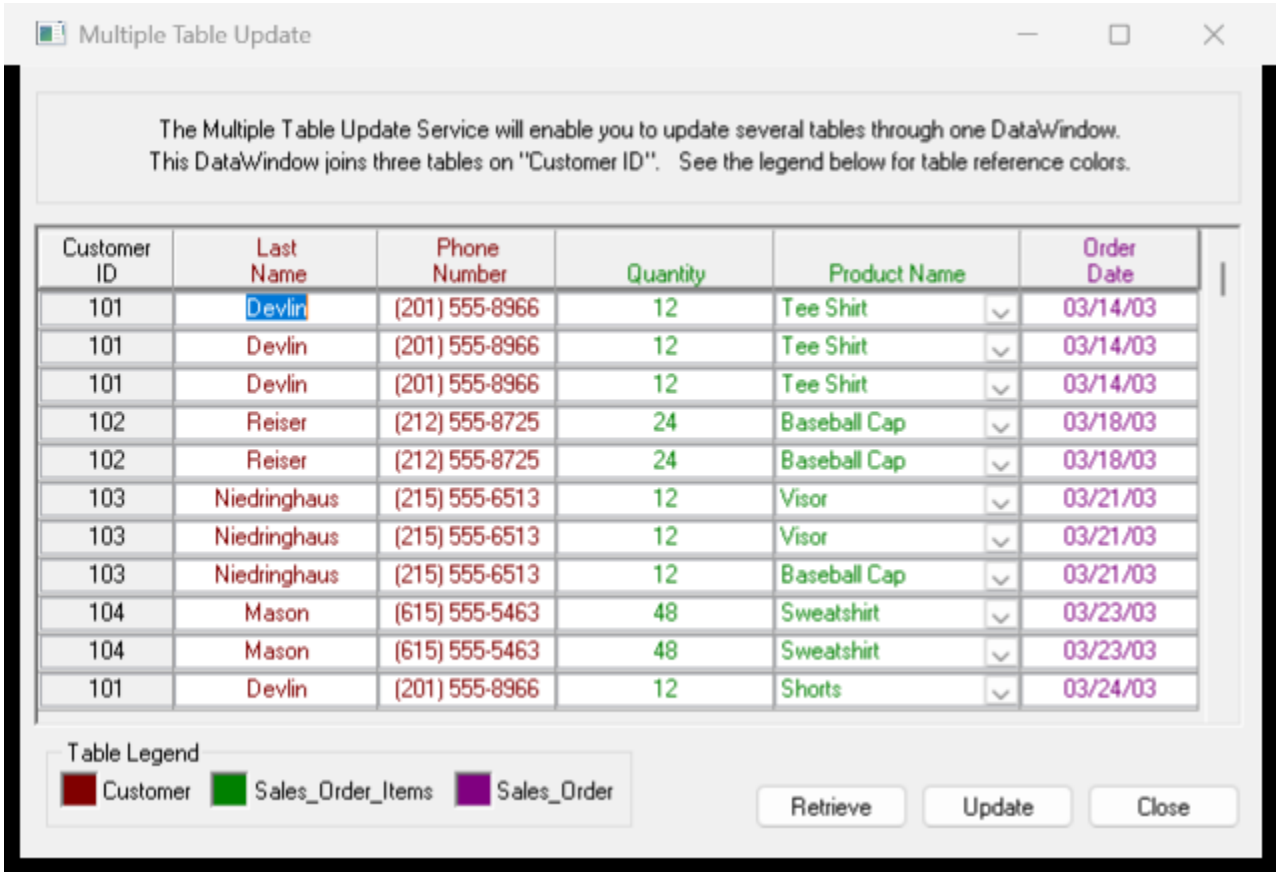
`n_cst_dwsrv_grid`, `w_grid`

The window

`w_gridserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Multiple Table Update

The multiple table update service (`n_cst_dwsrv_multitable`) updates several tables from one DataWindow: here one that joins customer, sales_order and sales_order_items, with a color for each table.



Try it

- Retrieve, change values from each table, and click Update.

PFC objects

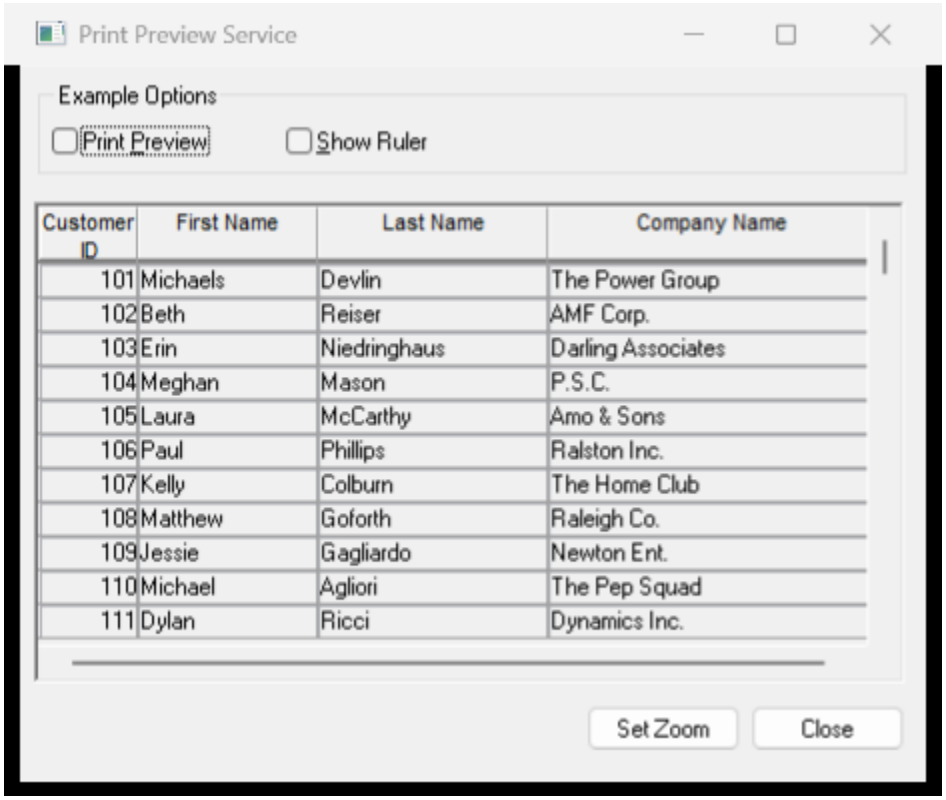
`n_cst_dwsrv_multitable`

The window

`w_multiupdate`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Print Preview Service

The print preview service (`n_cst_dwsrv_printpreview`) previews a DataWindow as it will print, with a ruler and a zoom dialog box.



Try it

- Click Print Preview, check Show Ruler, and click Set Zoom.

PFC objects

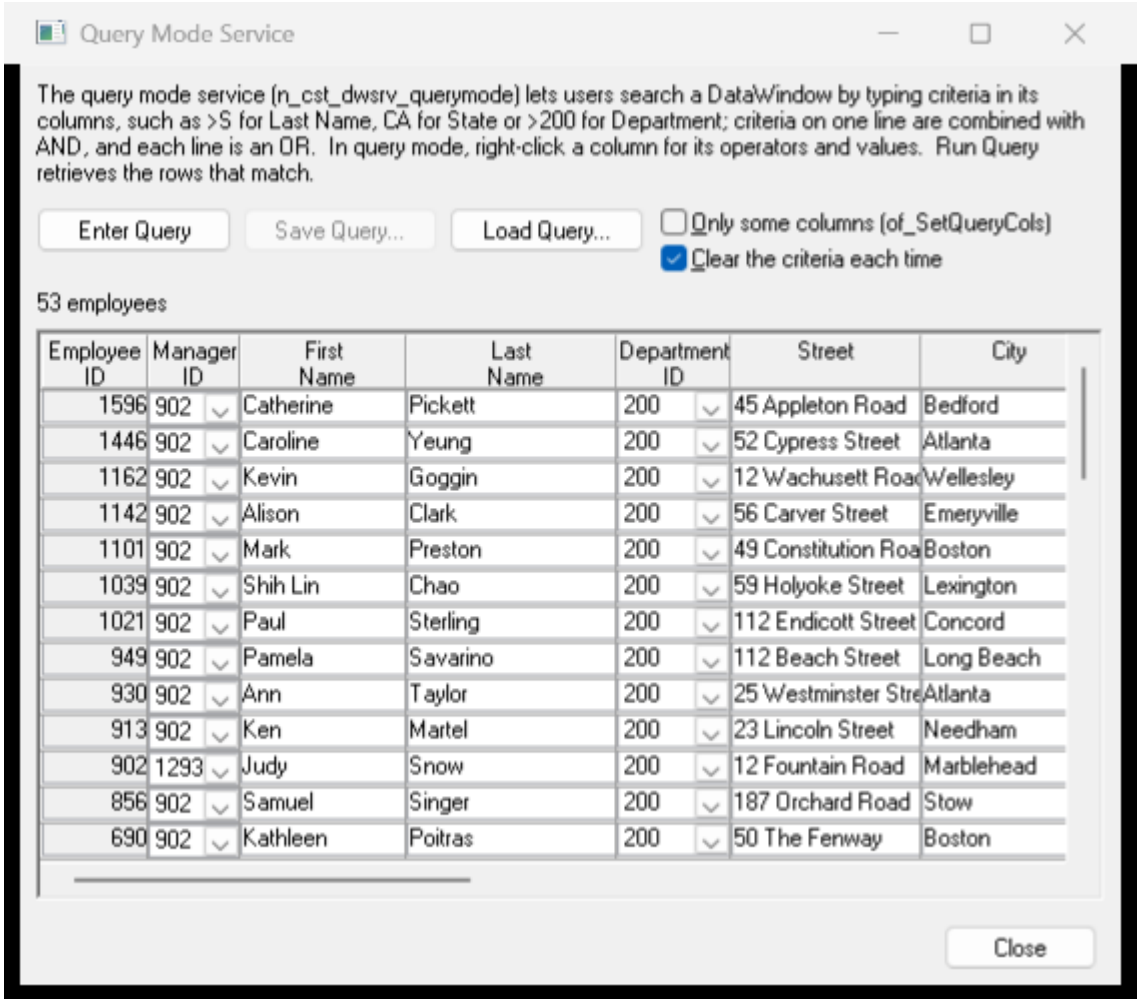
`n_cst_dwsrv_printpreview`

The window

`w_dwprintpreview`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Query Mode Service

The query mode service (`n_cst_dwsrv_querymode`) lets users search a DataWindow by typing criteria in its columns, such as >S or CA; criteria on one line are combined with AND, and each line is an OR. Queries can be saved and loaded.



Try it

- Turn query mode on, type criteria, and retrieve.
- In query mode, right-click a column for its operators and values; save the query and load it again.

PFC objects

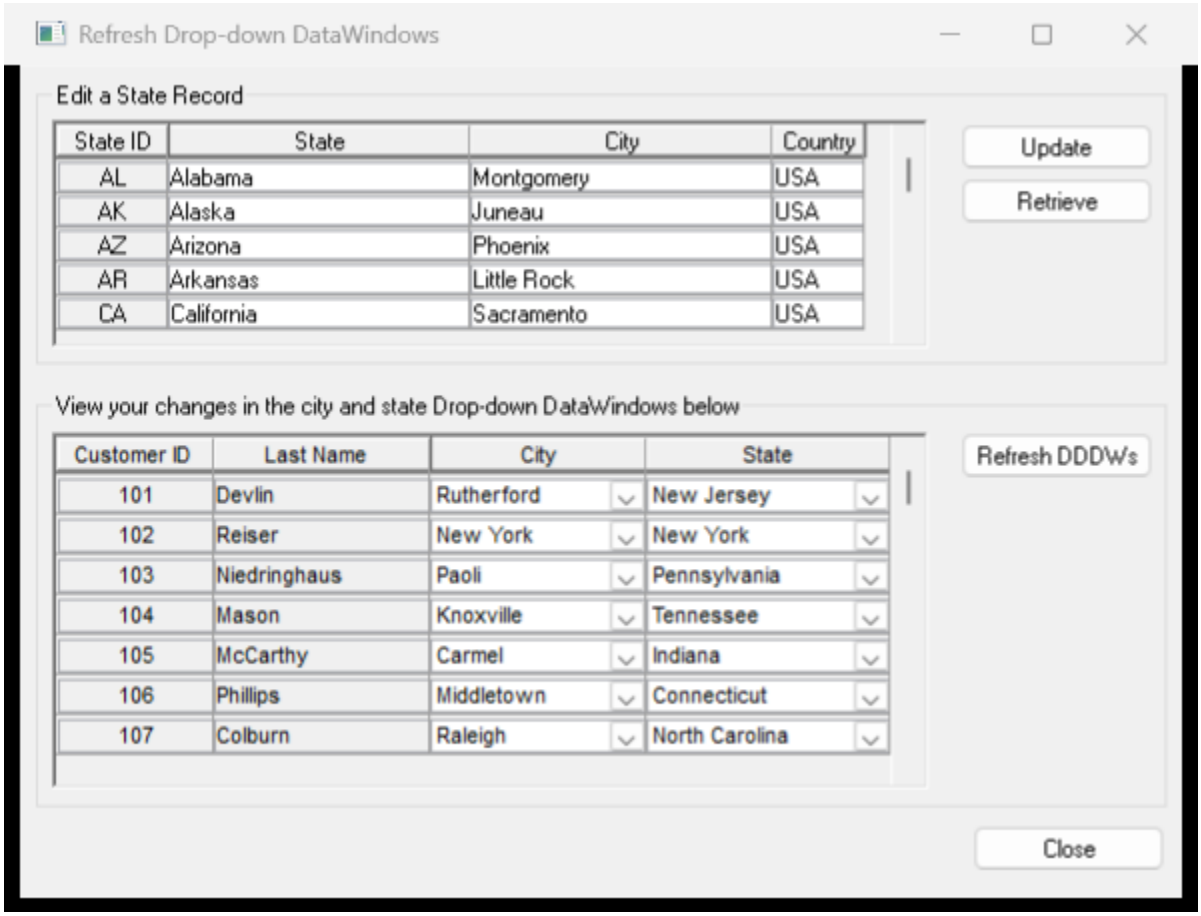
`n_cst_dwsrv_querymode`

The window

`w_querymode`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Refresh Drop-down DataWindows

The base DataWindow service's `of_RefreshDDDWs` retrieves a DataWindow's drop-down DataWindows again, so they show the data's changes.



Try it

- Edit a state, click Update, then Refresh DDDWs: the city and state drop-down DataWindows below show the change.

PFC objects

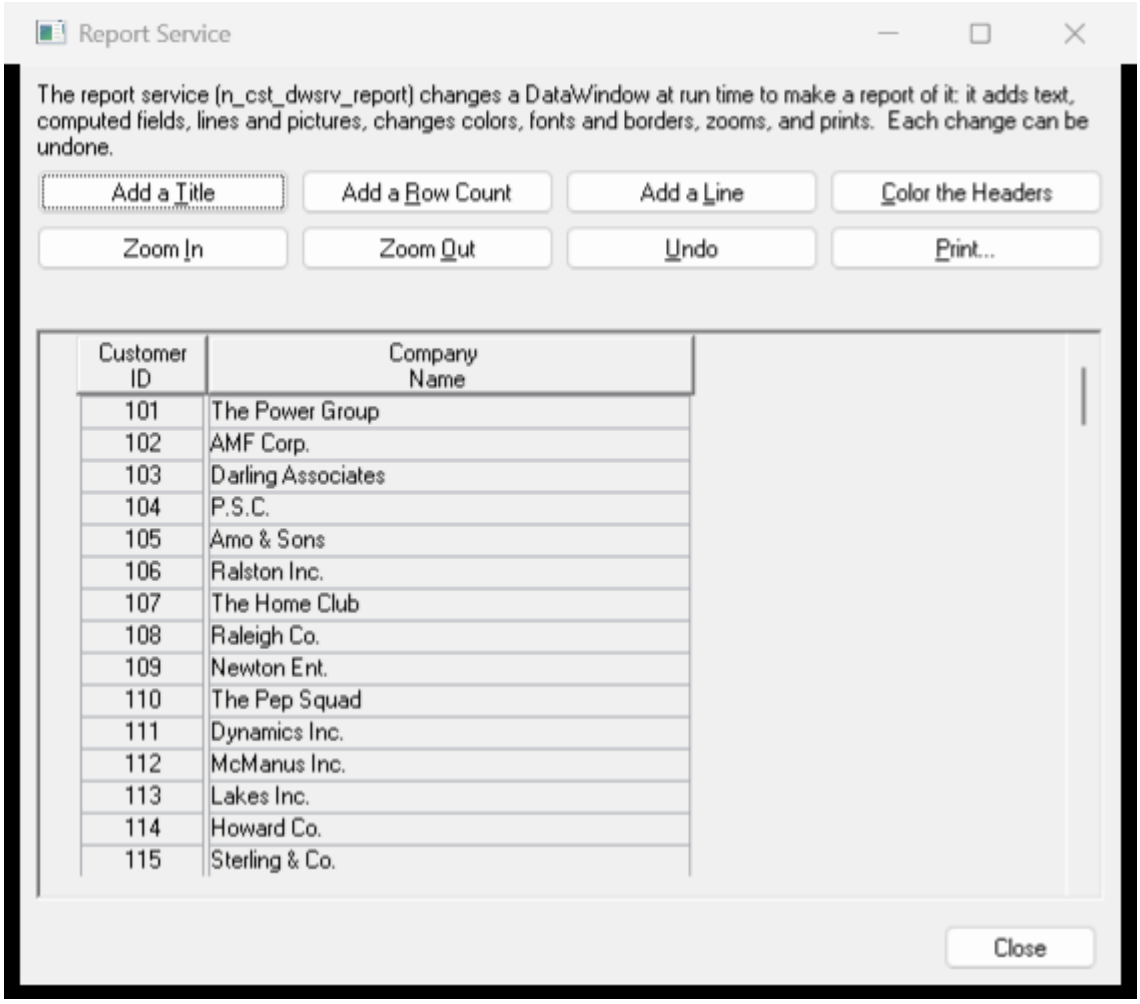
`n_cst_dwsrv`

The window

`w_refreshdddws`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Report Service

The report service (`n_cst_dwsrv_report`) changes a DataWindow at run time to make a report of it: it adds text, computed fields, lines and pictures, changes colors, fonts and borders, zooms and prints. Each change can be undone.



Try it

- Try the changes one at a time, and undo them.

PFC objects

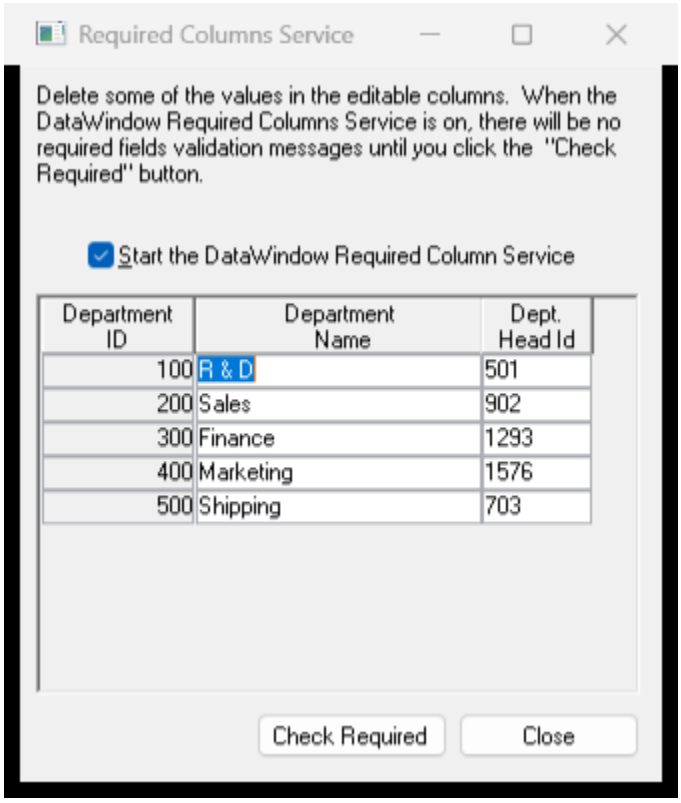
`n_cst_dwsrv_report`

The window

`w_reportserv`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Required Columns Service

The required columns service (`n_cst_dwsrv_reqcolumn`) leaves the check of required columns until the save, instead of when the user leaves a column, so the user can move about freely.



Try it

- Delete some values in the editable columns, move to other columns, then save.

PFC objects

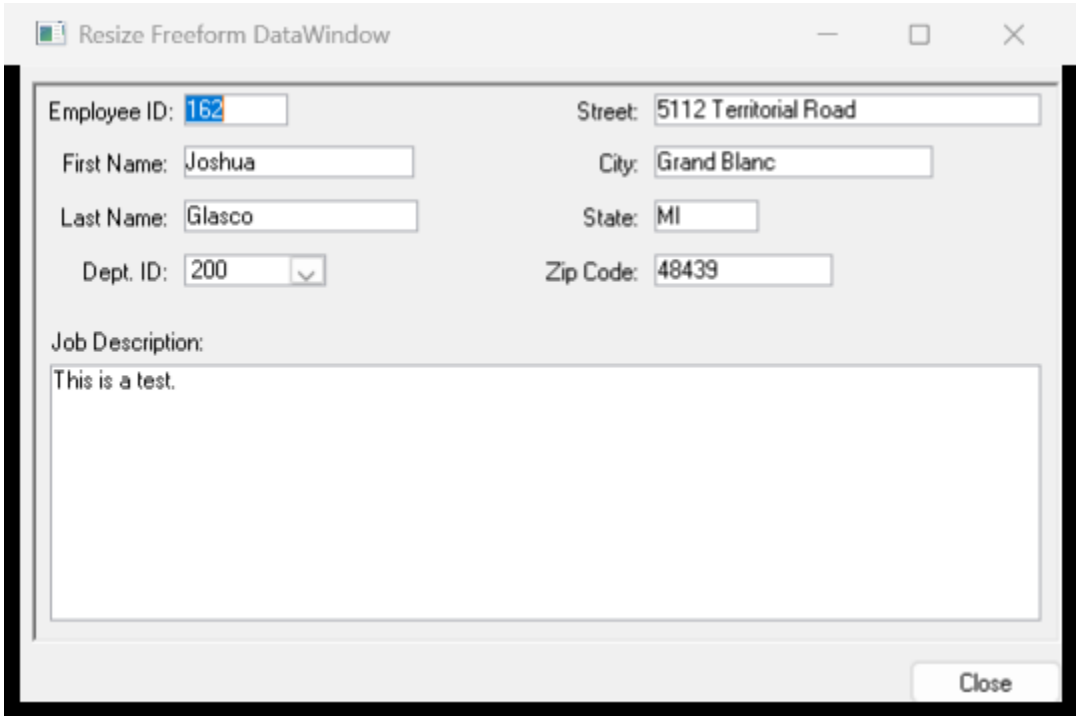
`n_cst_dwsrv_reqcolumn`

The window

`w_requiredfields`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Resize Freeform DataWindow

The DataWindow resize service (`n_cst_dwsrv_resize`) moves and resizes a DataWindow's columns when the DataWindow is resized; here a freeform DataWindow.



Try it

- Resize the window.

PFC objects

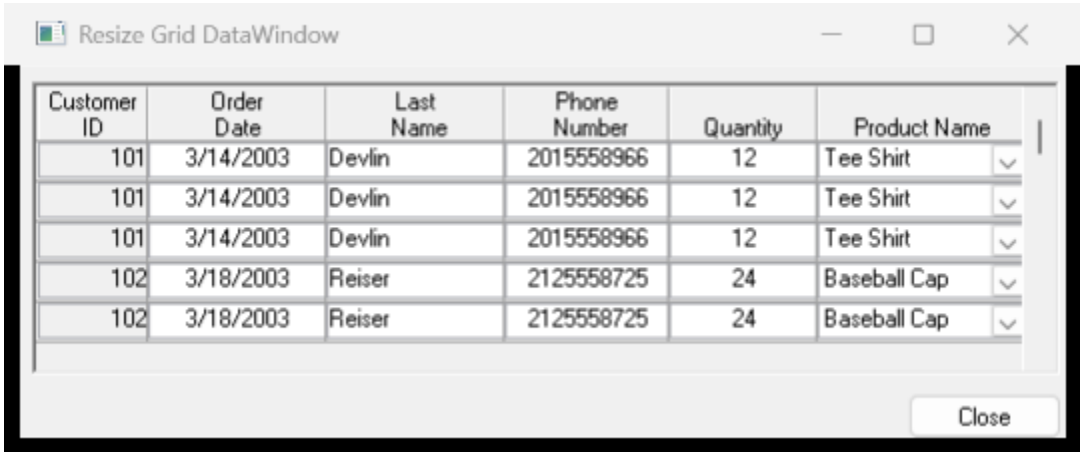
`n_cst_dwsrv_resize`

The window

`w_dwresizeff`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Resize Grid DataWindow

The DataWindow resize service with a grid DataWindow: the columns grow and shrink with the window.



Try it

- Resize the window.

PFC objects

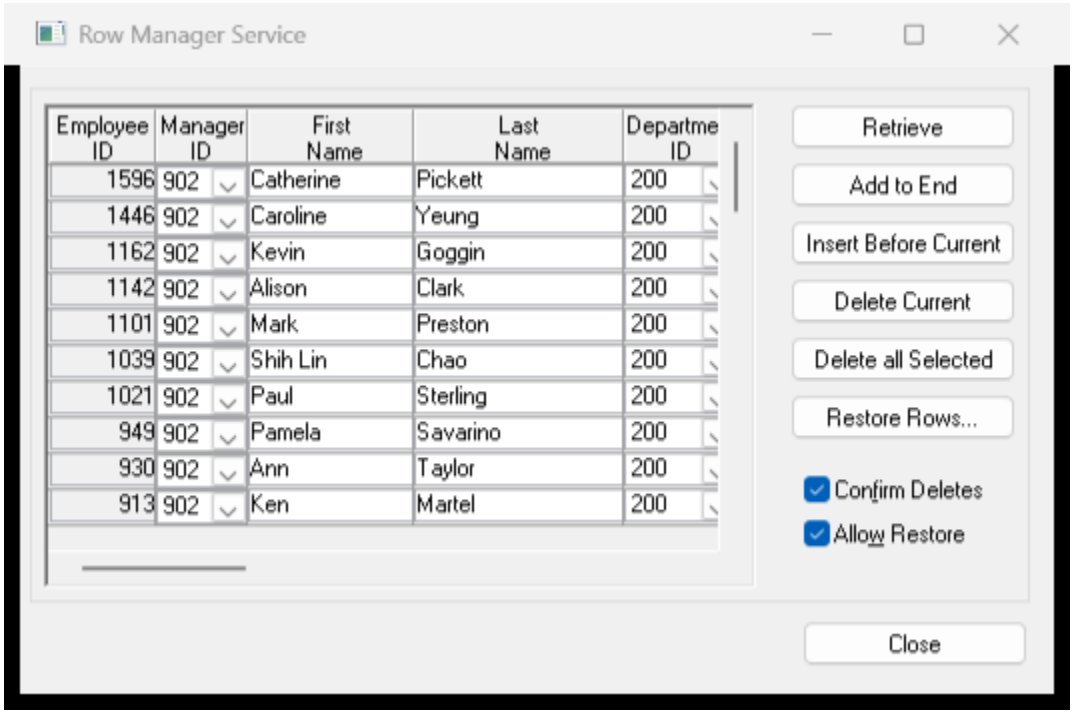
n_cst_dwsrv_resize

The window

`w_dwresizegrid`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Row Manager Service

The row manager service (`n_cst_dwsrv_rowmanager`) adds, inserts and deletes rows, asks before deleting, and lets the user restore deleted rows.



Try it

- Add a row at the end, insert one before the current row, and delete the current or the selected rows.
- Check Allow Restore, delete some rows, and click Restore Rows to bring them back.

PFC objects

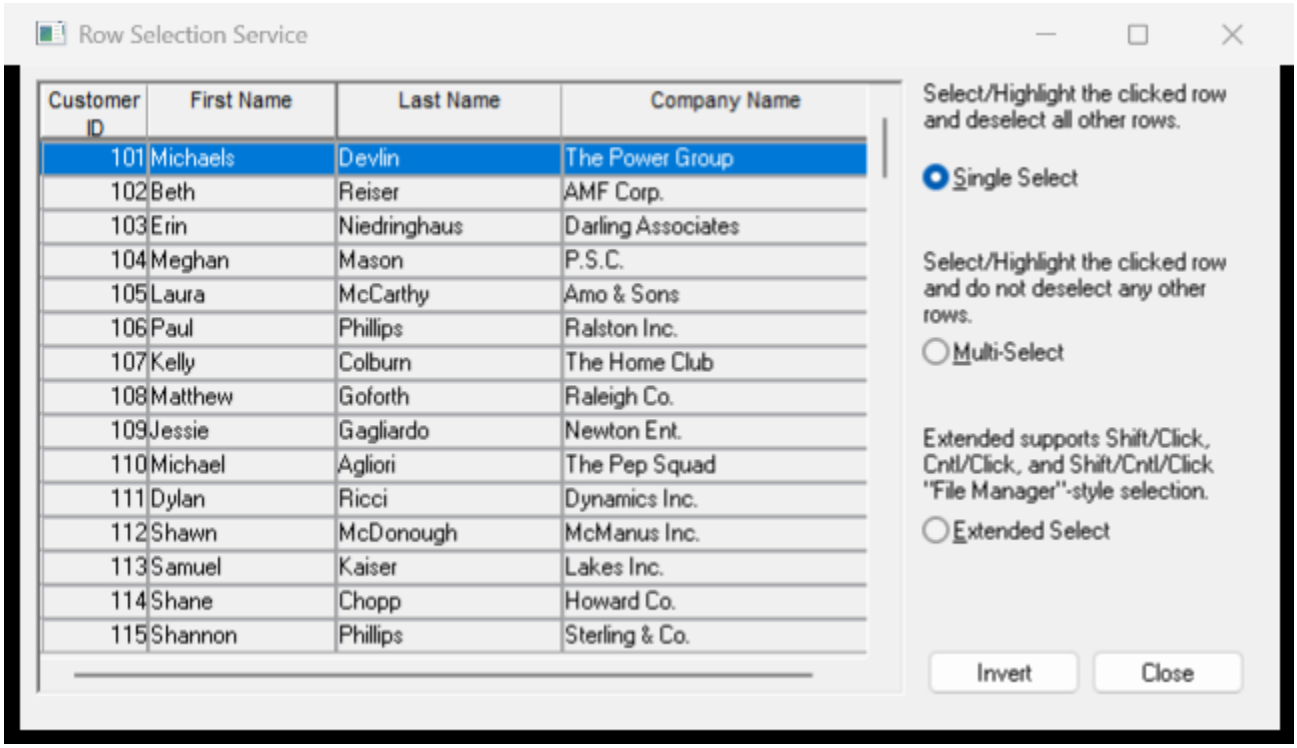
`n_cst_dwsrv_rowmanager`, `w_restorerow`

The window

`w_rowmanager`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Row Selection Service

The row selection service (`n_cst_dwsrv_rowselection`) highlights the rows the user clicks: one row at a time (single), each click adding a row (multi), or extended selection with Shift and Ctrl.



Try it

- Choose Single Select, Multi-Select or Extended Select, and click the rows.
- Click Invert to invert the selection.

PFC objects

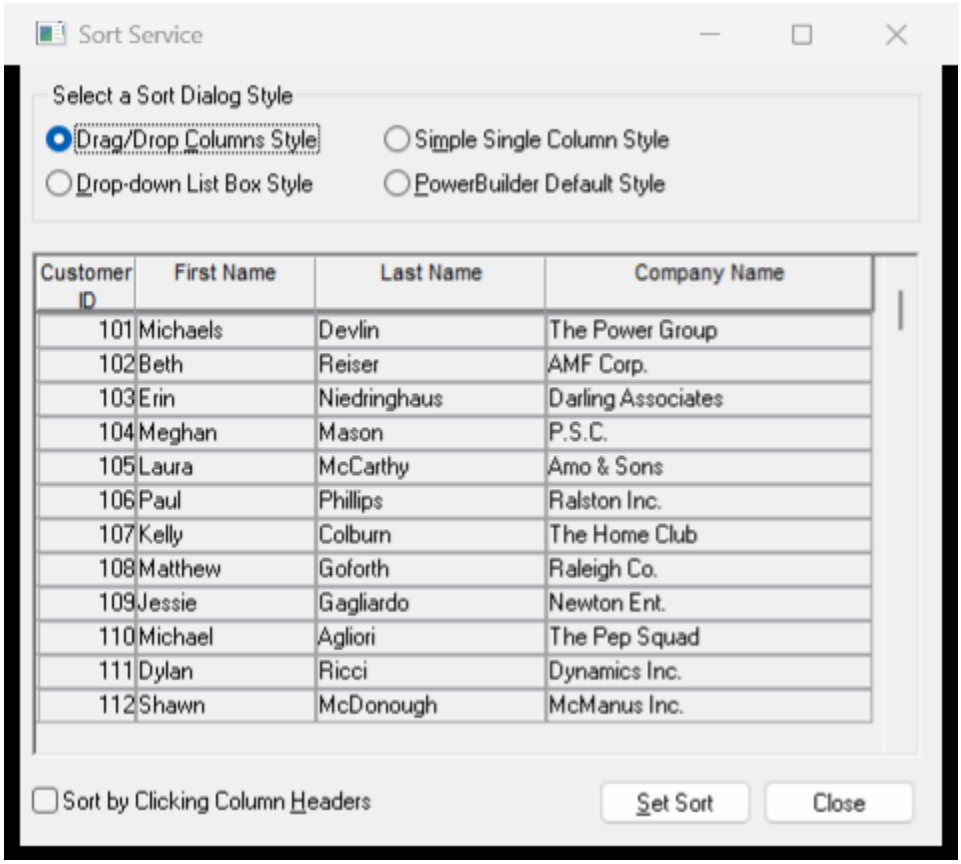
`n_cst_dwsrv_rowselection`

The window

`w_rowselection`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Sort Service

The sort service (`n_cst_dwsrv_sort`) shows one of four sort dialog boxes, and can sort a DataWindow when the user clicks a column header.



Try it

- Choose a dialog style (PowerBuilder's, drag and drop, drop-down list or a single column) and click Set Sort.
- Check Sort by Clicking Column Headers, then click the headers.

PFC objects

`n_cst_dwsrv_sort`

The window

`w_sort`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

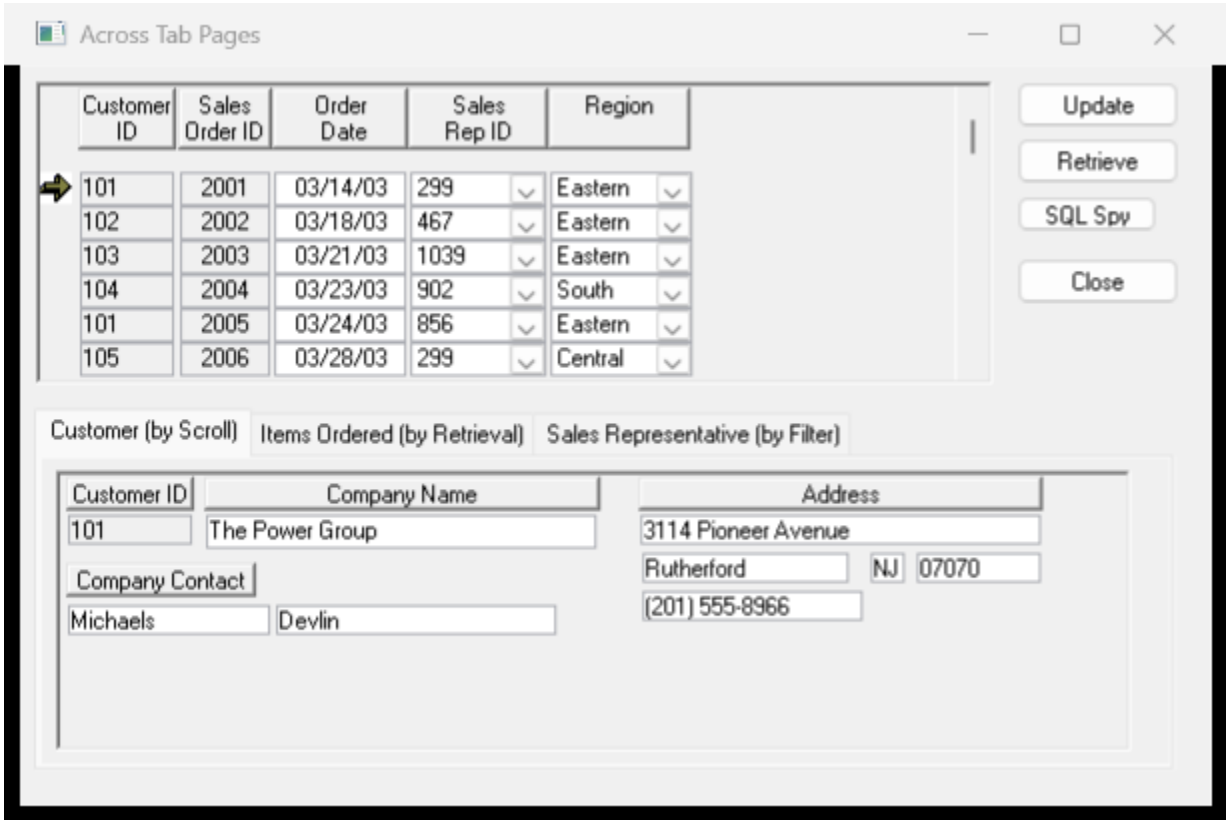
Linkage Services

The linkage service (`n_cst_dwsrv_linkage`) links DataWindows in master-detail relationships: by retrieving the details with retrieval arguments, by filtering them, or by scrolling to them.

Example	Window
Across Tab Pages	`w_tablink`
Basic Filters	`w_linkagefilter`
Basic Retrieval Arguments	`w_linkageretrieve`
Basic Scrolling	`w_linkagescroll`
Filter with Cascading Deletes	`w_linkcascadedeletes`
Filter with Cascading Keys	`w_linkcascadekeys`
Retrieval with Auto-updates	`w_linkautoupdate`

Across Tab Pages

The three linkage styles in one window: a master linked to three details, on tab pages, each with a different style.



Try it

- Retrieve, change the master row, and look at each tab page.

PFC objects

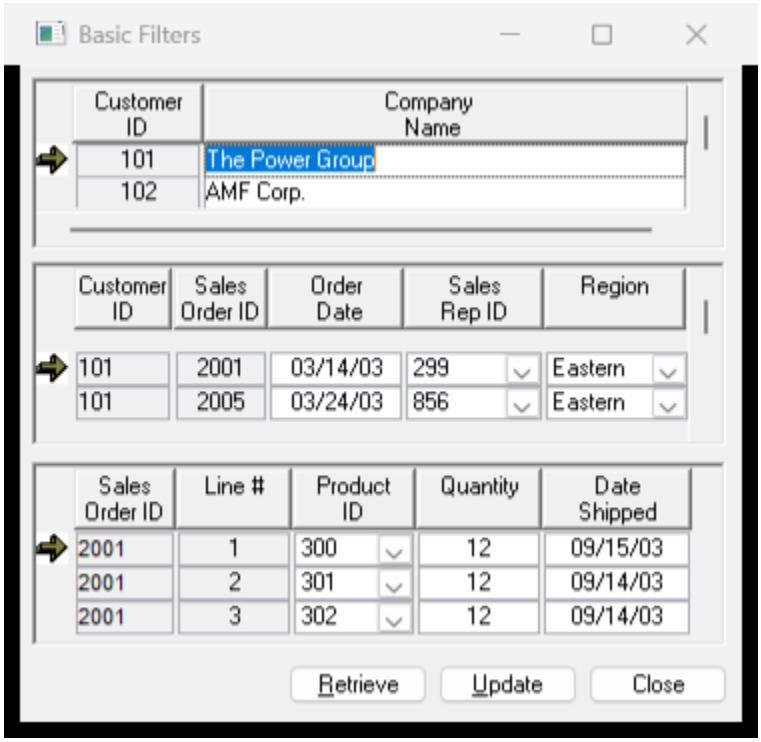
`n_cst_dwsrv_linkage`, `u_tab`

The window

`w_tablink`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Basic Filters

The filter style: all the rows are retrieved at once, and the details are filtered to the rows of the master's current row.



Try it

- Retrieve, then scroll through the master DataWindow; add and delete detail rows, and update.

PFC objects

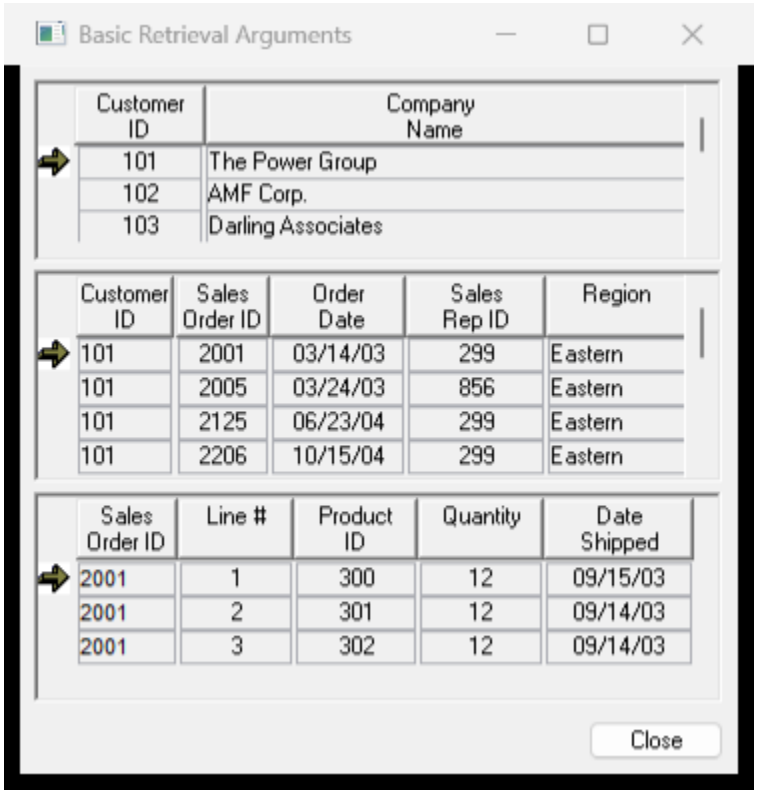
`n_cst_dwsrv_linkage`, `n_cst_dwsrv_rowmanager`

The window

`w_linkagefilter`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Basic Retrieval Arguments

The retrieve style: when the master's row changes, the details are retrieved with the master row's values as retrieval arguments.



Try it

- Scroll through the master DataWindow.

PFC objects

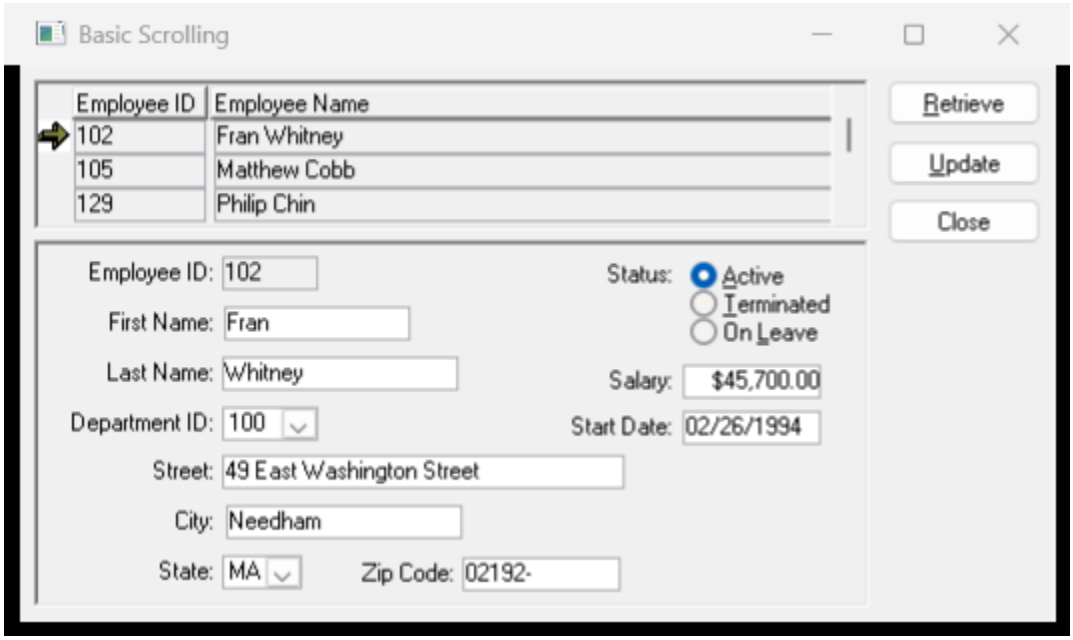
`n_cst_dwsrv_linkage`

The window

`w_linkageretrieve`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Basic Scrolling

The scroll style: all the rows are retrieved at once, and the details scroll to the rows of the master's current row.



Try it

- Retrieve, then scroll through the master DataWindow.

PFC objects

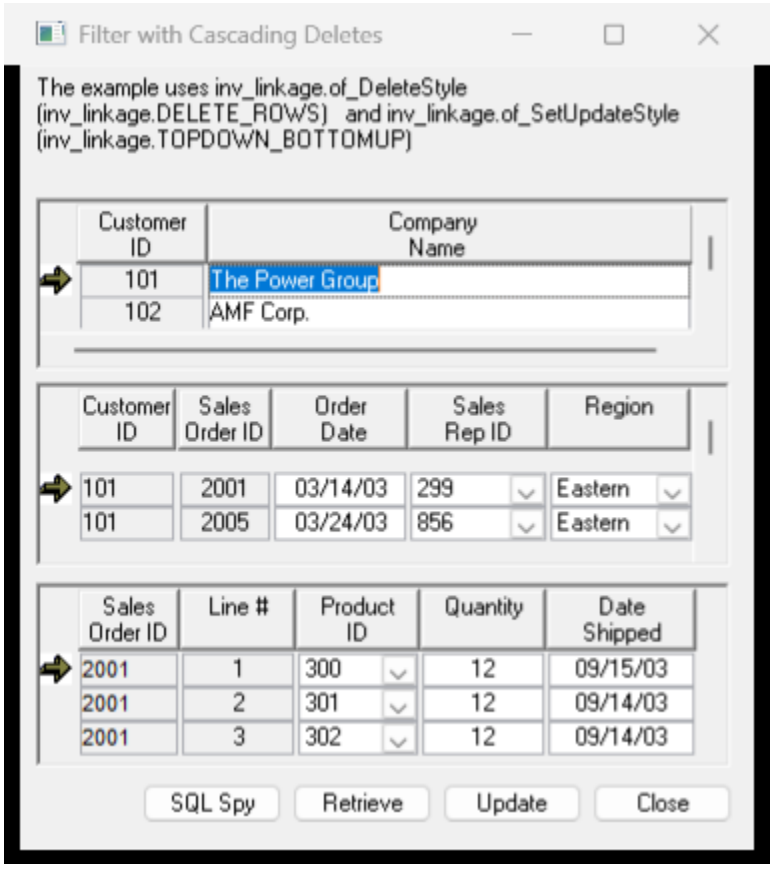
`n_cst_dwsrv_linkage`

The window

`w_linkagescroll`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Filter with Cascading Deletes

Deleting a master row deletes its details too (`of_SetDeleteStyle`), and the update runs in the right order (`of_SetUpdateStyle`).



Try it

- Delete a master row, and save.

PFC objects

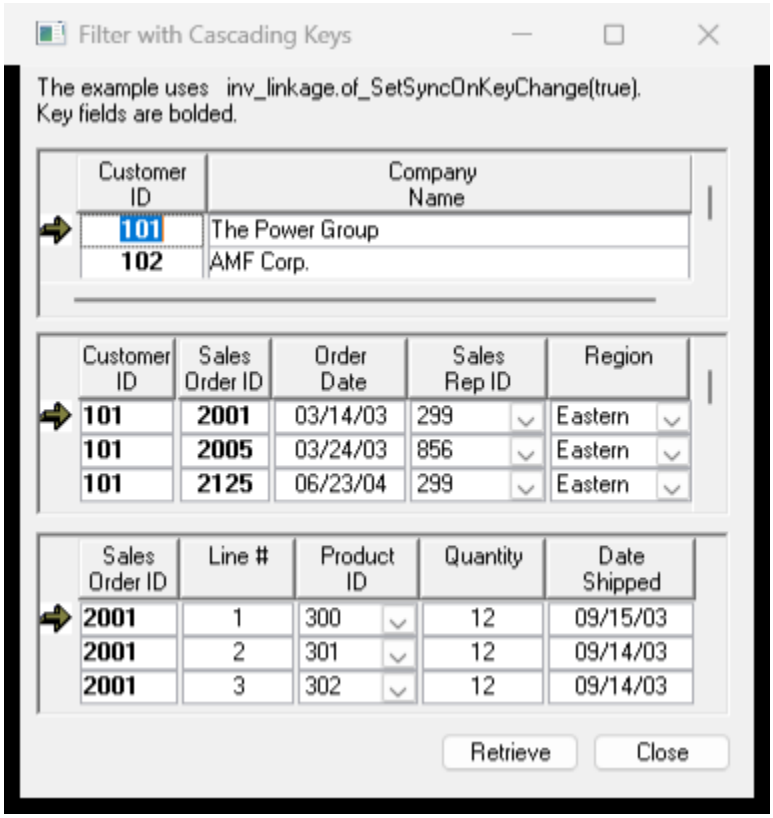
`n_cst_dwsrv_linkage`

The window

`w_linkcascadedeletes`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Filter with Cascading Keys

When a master's key changes, the linkage service changes it in the details too (`of_SetSyncOnKeyChange`).



Try it

- Change a key column (in bold) in the master, and see the details follow.

PFC objects

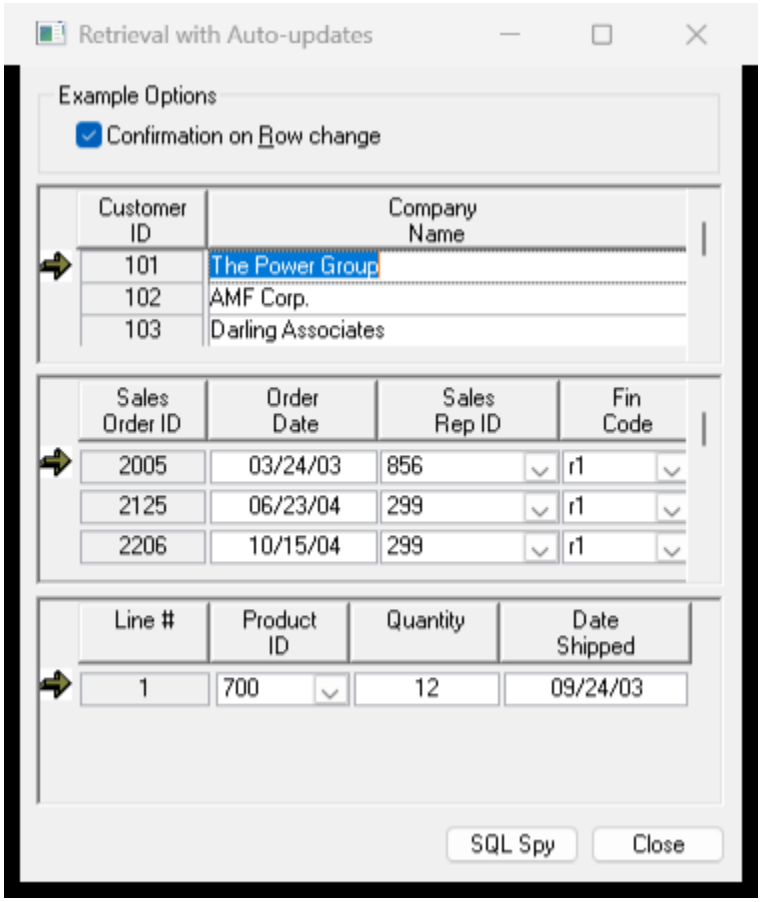
`n_cst_dwsrv_linkage`

The window

`w_linkcascadekeys`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Retrieval with Auto-updates

The linkage service can save the details when the master row changes (`of_SetUpdateOnRowChange`), and ask first (`of_SetConfirmOnRowChange`).



Try it

- Change a detail, then move to another master row.
- Check Confirmation on Row Change, and open the SQL Spy to see the updates.

PFC objects

`n_cst_dwsrv_linkage`

The window

`w_linkautoupdate`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

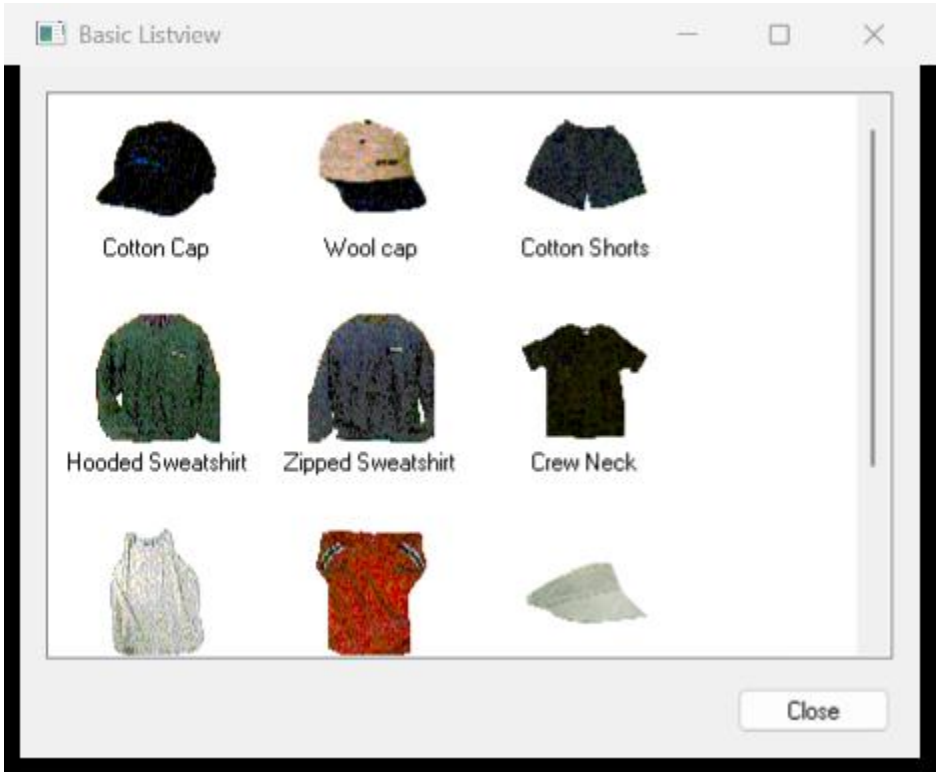
TreeView and Listview

PFC's TreeView and ListView controls with services (`u_tvs` and `u_lvs`) fill themselves from DataWindow objects, DataStores or SQL, one level at a time.

Example	Window
Basic Listview	`w_listview6`
Basic Treeview	`w_treeview6`
Recursive Tree View	`w_treeviewrecursive6`
Treeview Linked to a Listview	`w_treeviewlistview6`
Update a Treeview	`w_treeviewupdate6`

Basic Listview

A ListView filled from a DataWindow object by its data source service, with report columns, pictures and sorting.



Try it

- Change the view, and click a column header in the report view to sort.

PFC objects

`u_lvs`, `n_cst_lvsrv_datasource`

The window

`w_listview6`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Basic Treeview

A TreeView with six levels, from sales regions down to the items of an order, each from its own DataWindow object (the level source service, `n_cst_tvsrv_levelsource`), linked to a DataWindow that shows the selected item.



Try it

- Expand the levels, and select items to see them in the DataWindow; change one and save.

PFC objects

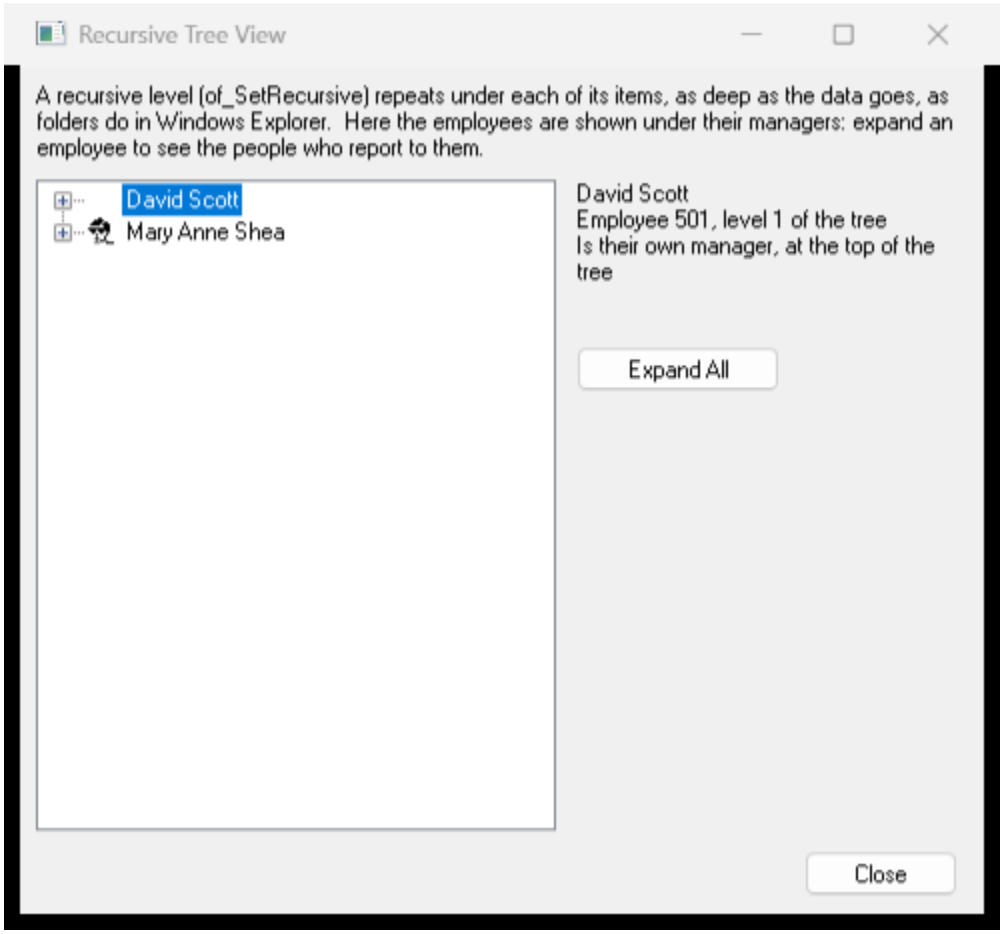
`u_tvs`, `n_cst_tvsrv_levelsource`

The window

`w_treeview6`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Recursive Tree View

A recursive level (`of_SetRecursive`) repeats under each of its items, as deep as the data goes, as folders do in Windows Explorer: here the employees are shown under their managers.



Try it

- Expand an employee to see the people who report to them, or click Expand All.

PFC objects

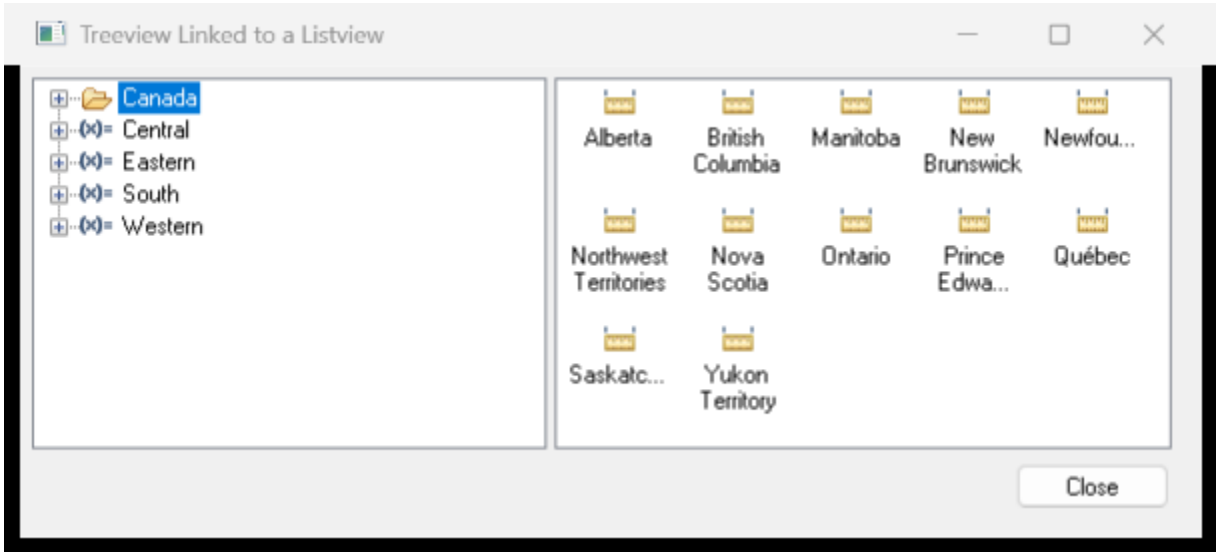
`u_tvs`, `n_cst_tvsrv_levelsource`

The window

`w_treeviewrecursive6`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Treeview Linked to a Listview

A TreeView linked to a ListView: the ListView (`u_lvs`, with its data source service) shows the items under the selected TreeView item.



Try it

- Select items in the TreeView, and change the ListView's view.

PFC objects

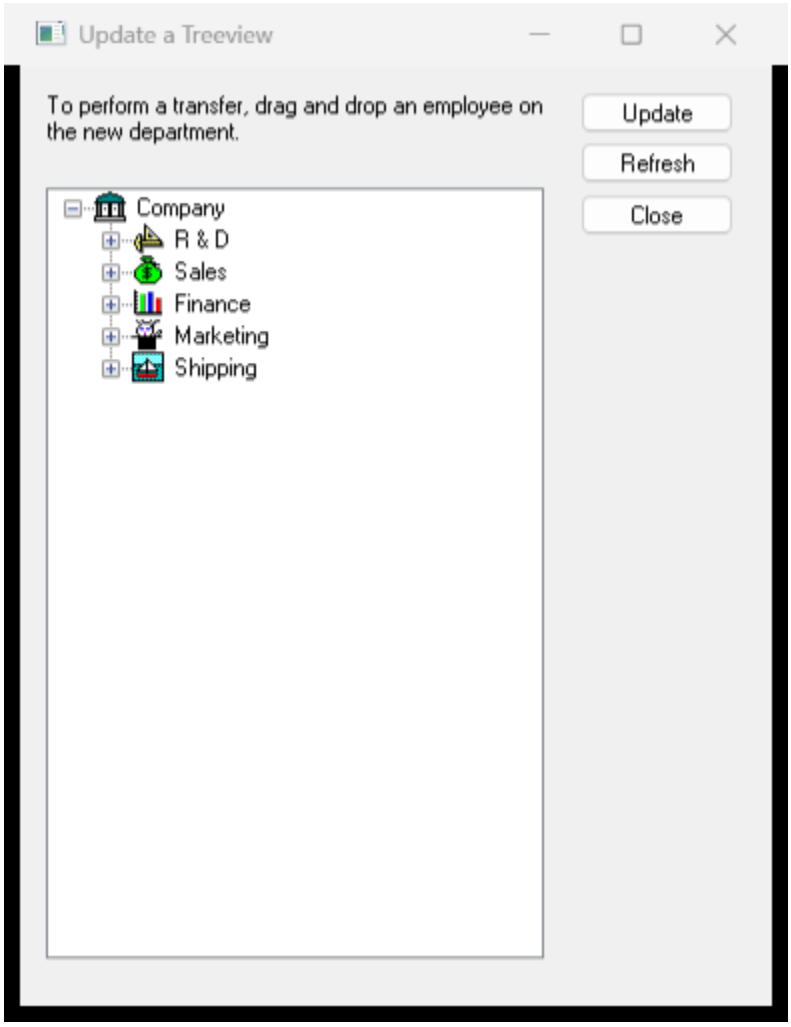
`u_tvs`, `u_lvs`, `n_cst_lvsrv_datasource`

The window

`w_treeviewlistview6`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Update a Treeview

The level source service updates the database from the TreeView: here employees are moved to another department by dragging them.



Try it

- Drag an employee onto another department.

PFC objects

`u_tvs`, `n_cst_tvsrv_levelsource`

The window

`w_treeviewupdate6`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

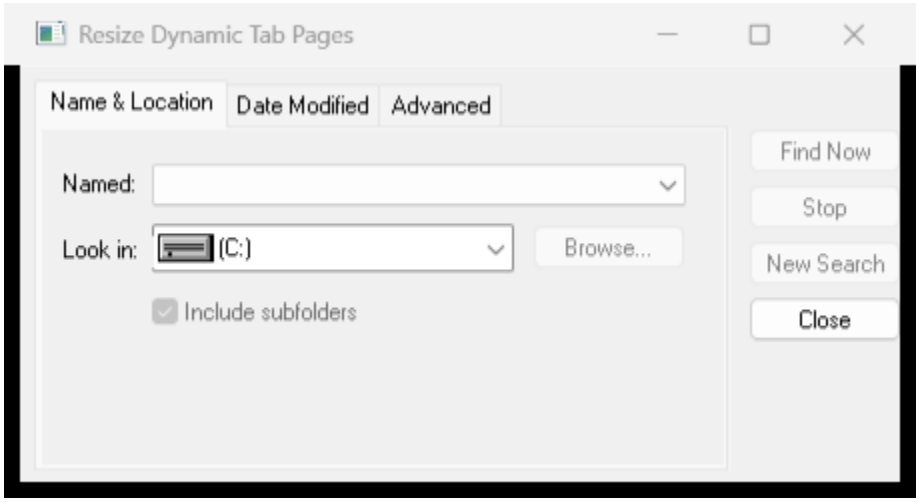
Window Services

The services of PFC windows: the save process, resizing, the status bar, the sheet manager, and a frame with a ribbon bar.

Example	Window
Resize Dynamic Tab Pages	`w_tabresize`
Ribbon Bar	`w_ribbonframe`
Save Process: All DW Scenarios	`w_dw3types`
Sheet Manager	`w_sheetframe`
Status Bar	`w_statusbarframe`
Weighted Resize of a Window	`w_resizedwsyntax`
Window Resize Service	`w_resizefind`

Resize Dynamic Tab Pages

The resize service on tab pages that are created when they are first selected.



Try it

- Resize the window, and select the tab pages.

PFC objects

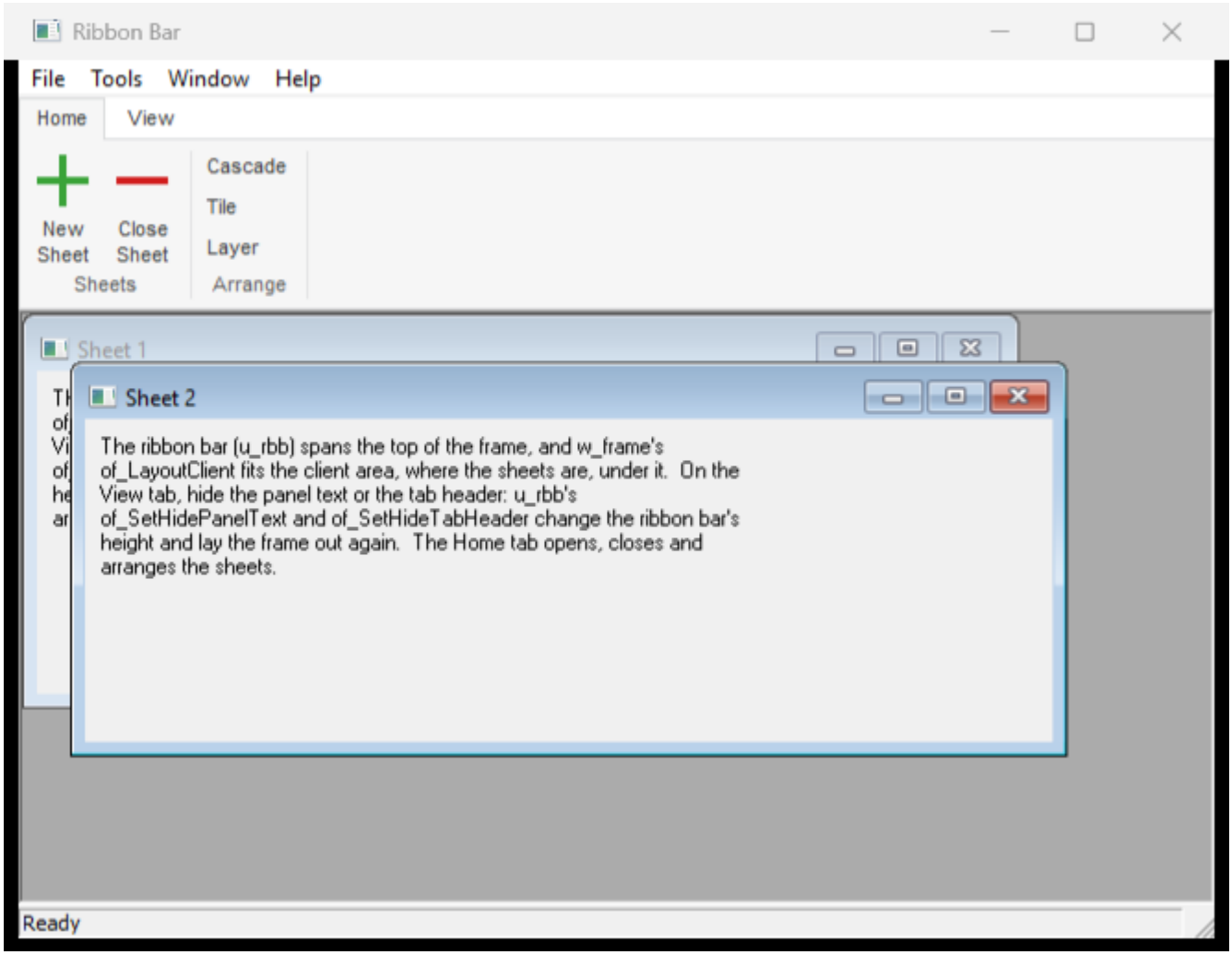
n_cst_resize, u_tab

The window

w_tabresize, in appexamp.pbl. Open it in the PowerBuilder IDE to see how the example is coded.

Ribbon Bar

A frame with a ribbon bar (`u_rbb`): `w_frame` lays out its client area under the ribbon bar, and again when `u_rbb`'s `of_SetHidePanelText` or `of_SetHideTabHeader` changes the ribbon bar's height.



Try it

- Open, close and arrange the sheets from the Home tab.
- On the View tab, hide the panel text or the tab header.

PFC objects

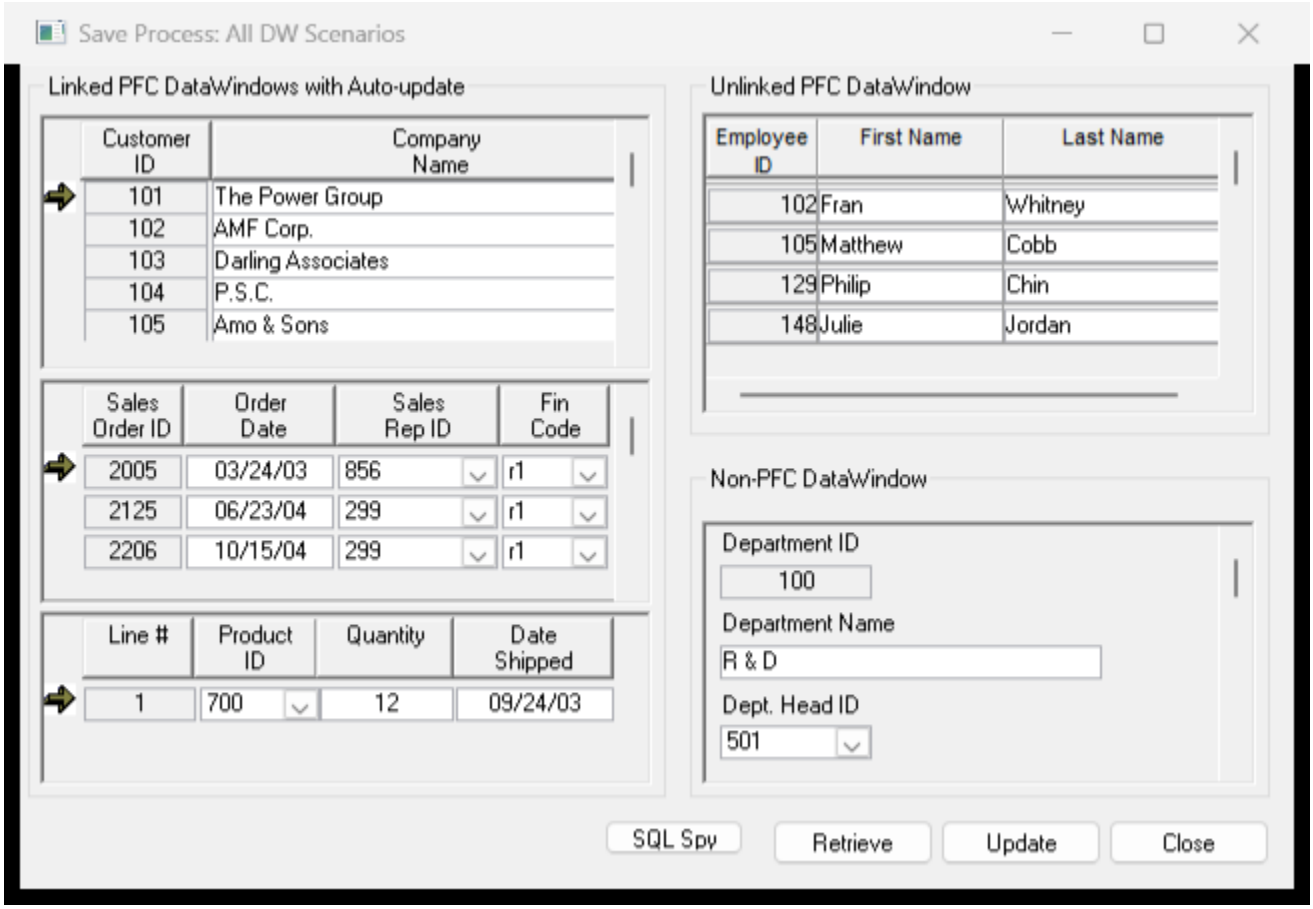
`u_rbb`, `w_frame`

The window

`w_ribbonframe`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Save Process: All DW Scenarios

The window save process (`pfc_save`, through `n_cst_luw`) with three kinds of DataWindows in one window: linked PFC DataWindows, an unlinked PFC DataWindow and a native PowerBuilder DataWindow. The SQL Spy shows the updates.



Try it

- Change data in each DataWindow and click Update.

PFC objects

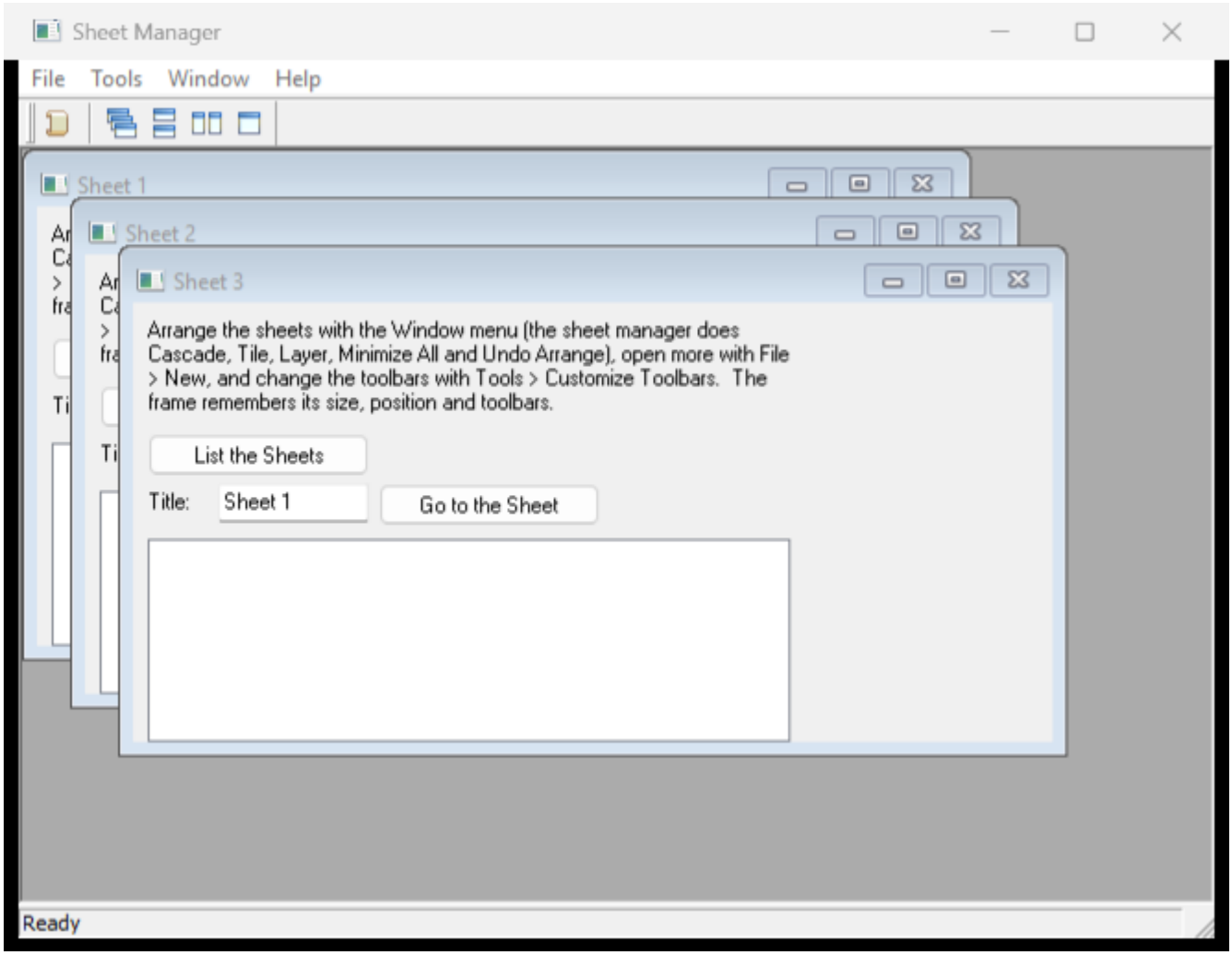
`w_master`, `n_cst_luw`, `n_cst_dwsrv_linkage`

The window

`w_dw3types`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Sheet Manager

The sheet manager (`n_cst_winsrv_sheetmanager`) arranges a frame's sheets (cascade, tile, layer, minimize all, undo arrange), and lists and finds them. The frame also keeps its size, position and toolbars.



Try it

- Arrange the sheets with the Window menu, open more with File > New, and list and find them on a sheet.

PFC objects

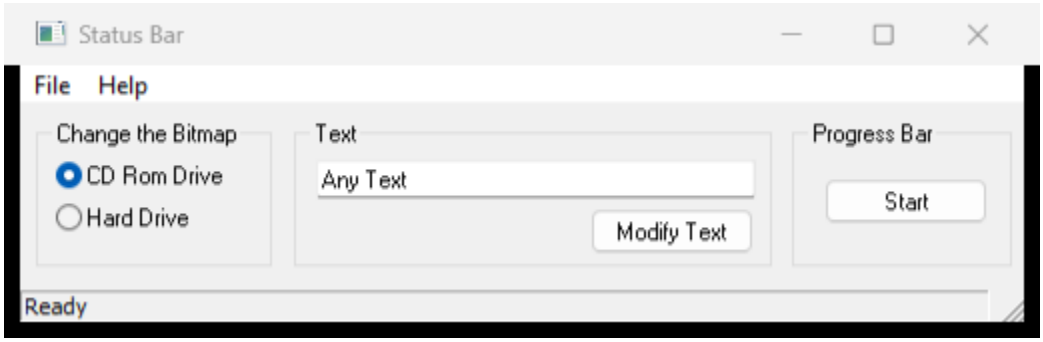
`n_cst_winsrv_sheetmanager`, `w_frame`

The window

`w_sheetframe`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Status Bar

The status bar service (`n_cst_winsrv_statusbar`) adds a status bar to a frame, with text, pictures, a progress bar, the time and the free memory.



Try it

- Change the text and the pictures, and start the progress bar.

PFC objects

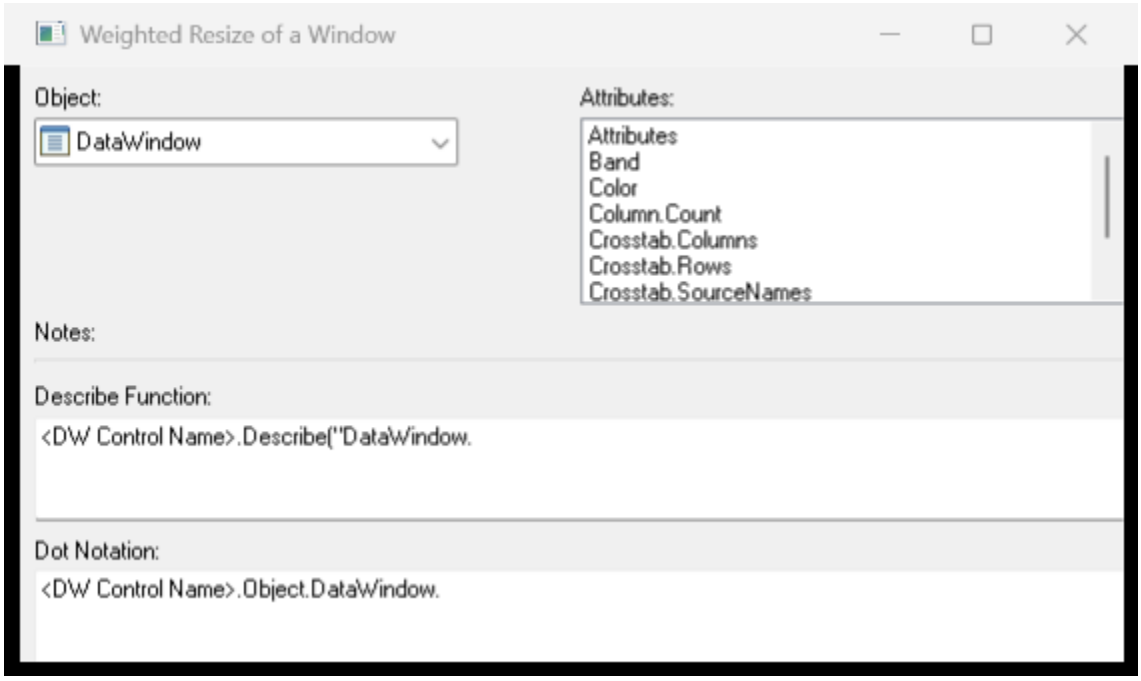
`n_cst_winsrv_statusbar`

The window

`w_statusbarframe`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Weighted Resize of a Window

Weighted resizing: the resize service shares a window's new size between its controls in the proportions you register.



Try it

- Resize the window.

PFC objects

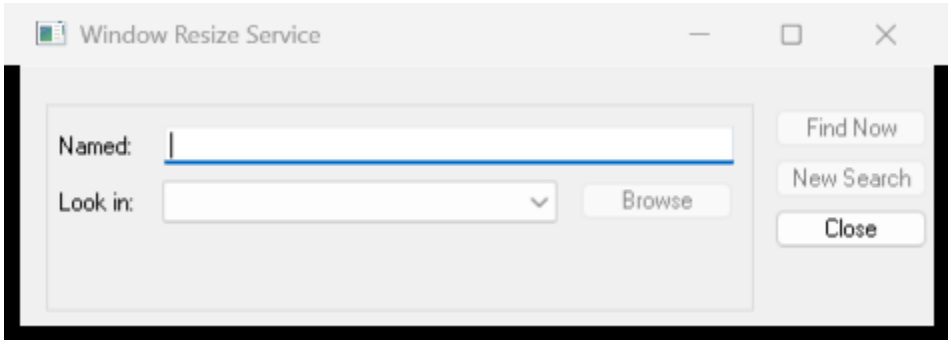
`n_cst_resize`

The window

`w_resizedwsyntax`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Window Resize Service

The window resize service (`n_cst_resize`) moves and resizes a window's controls when the window is resized, here in a window like Windows' Find dialog box, with a minimum size.



Try it

- Resize the window.

PFC objects

`n_cst_resize`

The window

`w_resizefind`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

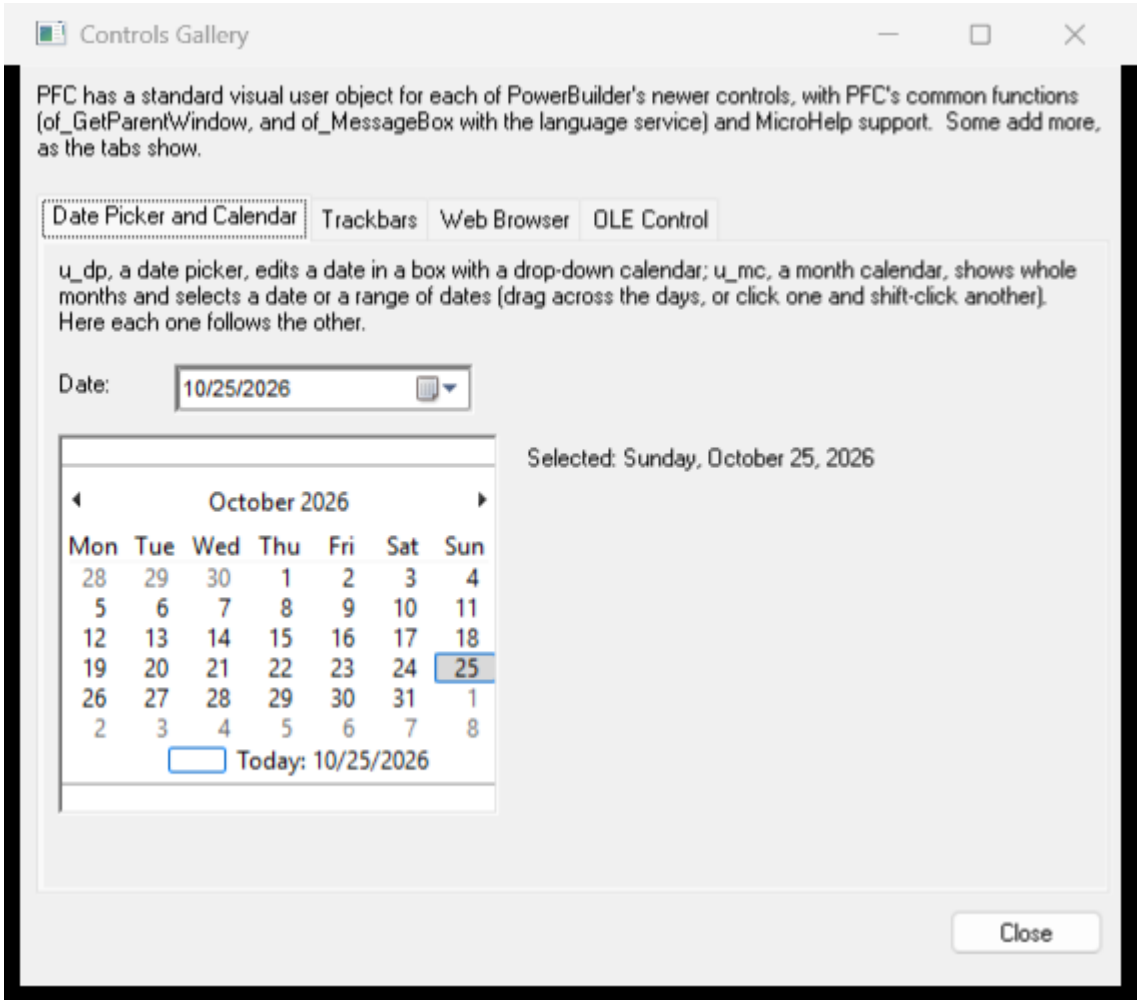
Objects

PFC's visual user objects: split bars, the progress bar, type-ahead list boxes, the Rich Text Edit control, and a gallery of the newer controls.

Example	Window
Controls Gallery	`w_controlgallery`
Find and Replace in a RTE	`w_findreplacerte`
Progress Bar Sampler	`w_progbarsampler`
Split Bar 3 Pane Style	`w_splitbar3panestyle`
Split Bar Explorer Style	`w_splitbarexplorerstyle`
Split Bar Settings	`w_splitbarwin`
Type Ahead in the DDLB & DDPLB	`w_dropdownsearchlb`

Controls Gallery

PFC's newer visual controls, a tab each: the date picker and month calendar (`u_dp`, `u_mc`), the trackbars (`u_htb`, `u_vtb`), the web browser and picture hyperlink (`u_wb`, `u_phl`) and the OLE control (`u_oc`).



Try it

- On each tab, use the controls; on the OLE Control tab, insert an object and right-click it.

PFC objects

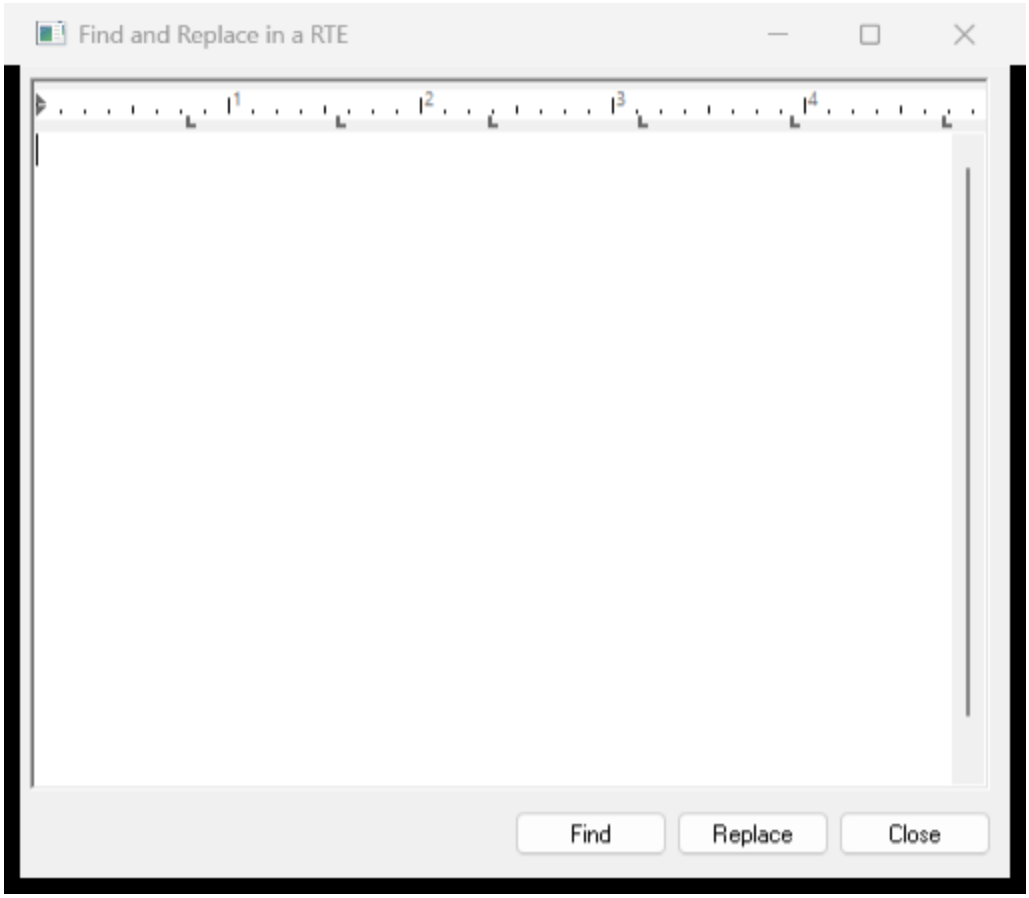
`u_dp`, `u_mc`, `u_htb`, `u_vtb`, `u_wb`, `u_phl`, `u_oc`

The window

`w_controlgallery`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Find and Replace in a RTE

The find service on PFC's Rich Text Edit control (`u_rte`).



Try it

- Click Find, or Replace, and search the text.

PFC objects

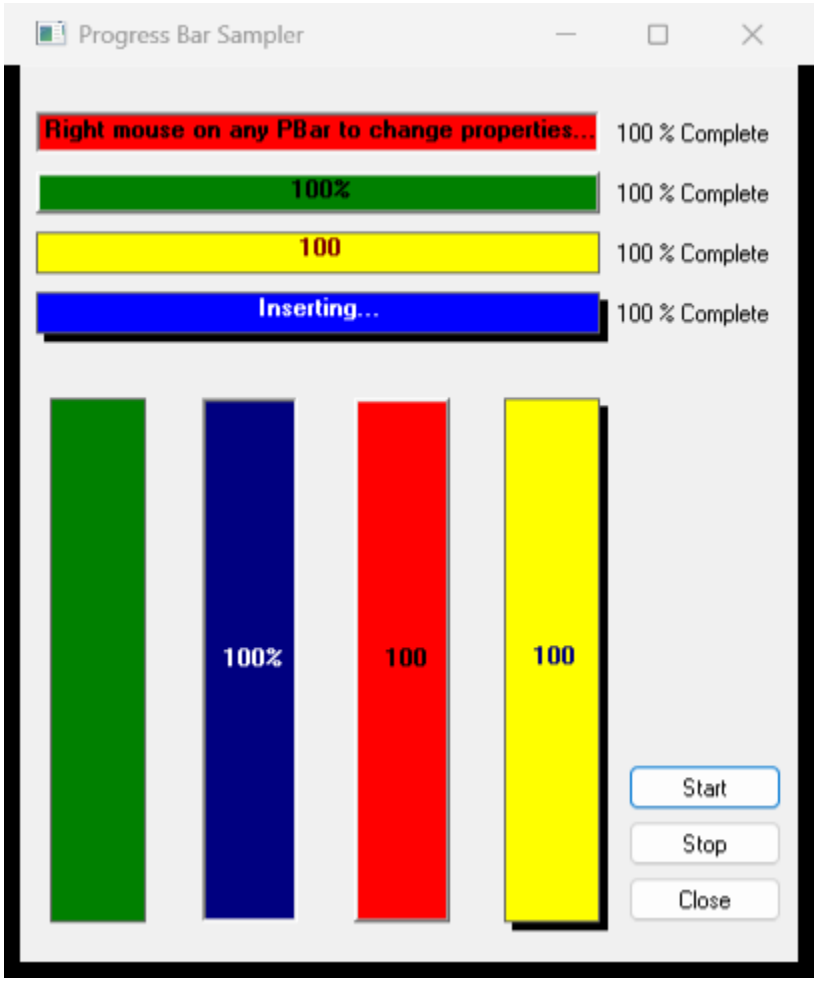
`u_rte`

The window

`w_findreplacerte`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Progress Bar Sampler

PFC's progress bar (u_progressbar) in its different styles: fill styles and colors, the text it displays, fonts, and resetting automatically.



Try it

- Click Start to run all the progress bars, and Stop.

PFC objects

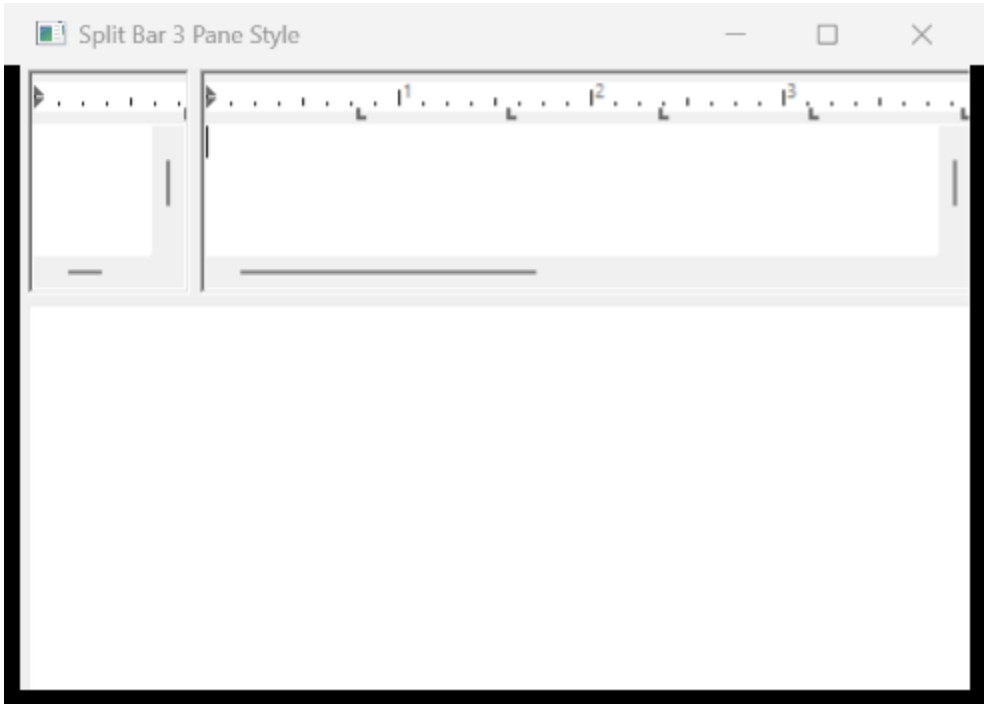
u_progressbar

The window

w_progbarsampler, in appexamp.pbl. Open it in the PowerBuilder IDE to see how the example is coded.

Split Bar 3 Pane Style

Split bars (`u_st_splitbar`) between three panes, one horizontal and one vertical, with the resize service.



Try it

- Drag the split bars, and resize the window.

PFC objects

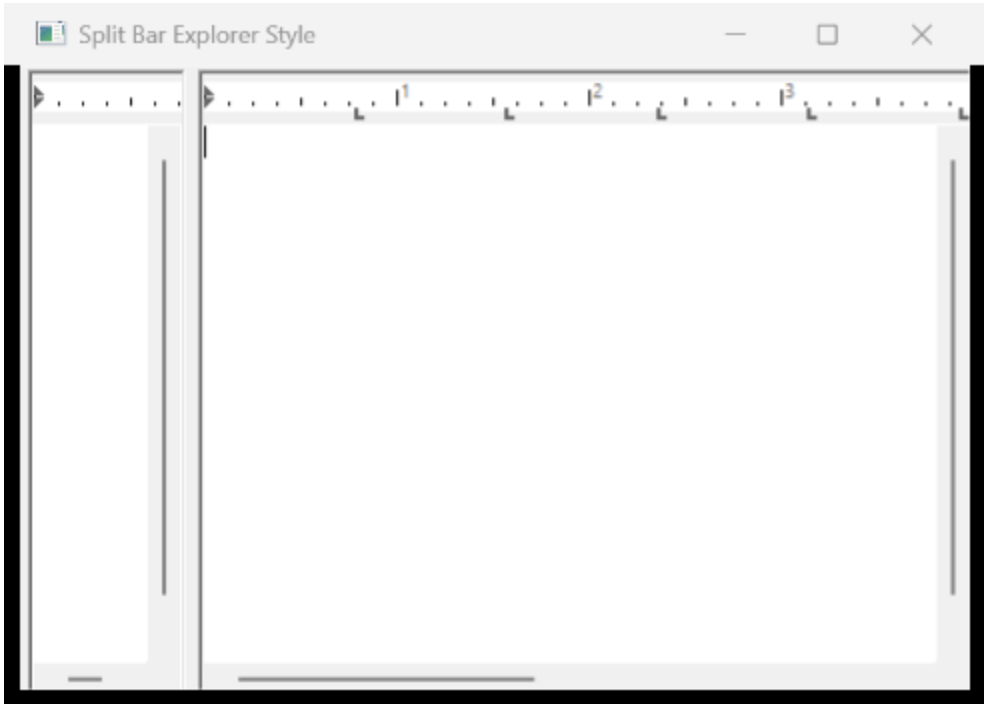
`u_st_splitbar`

The window

`w_splitbar3panestyle`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Split Bar Explorer Style

A split bar between two panes, as in Windows Explorer, with the resize service.



Try it

- Drag the split bar, and resize the window.

PFC objects

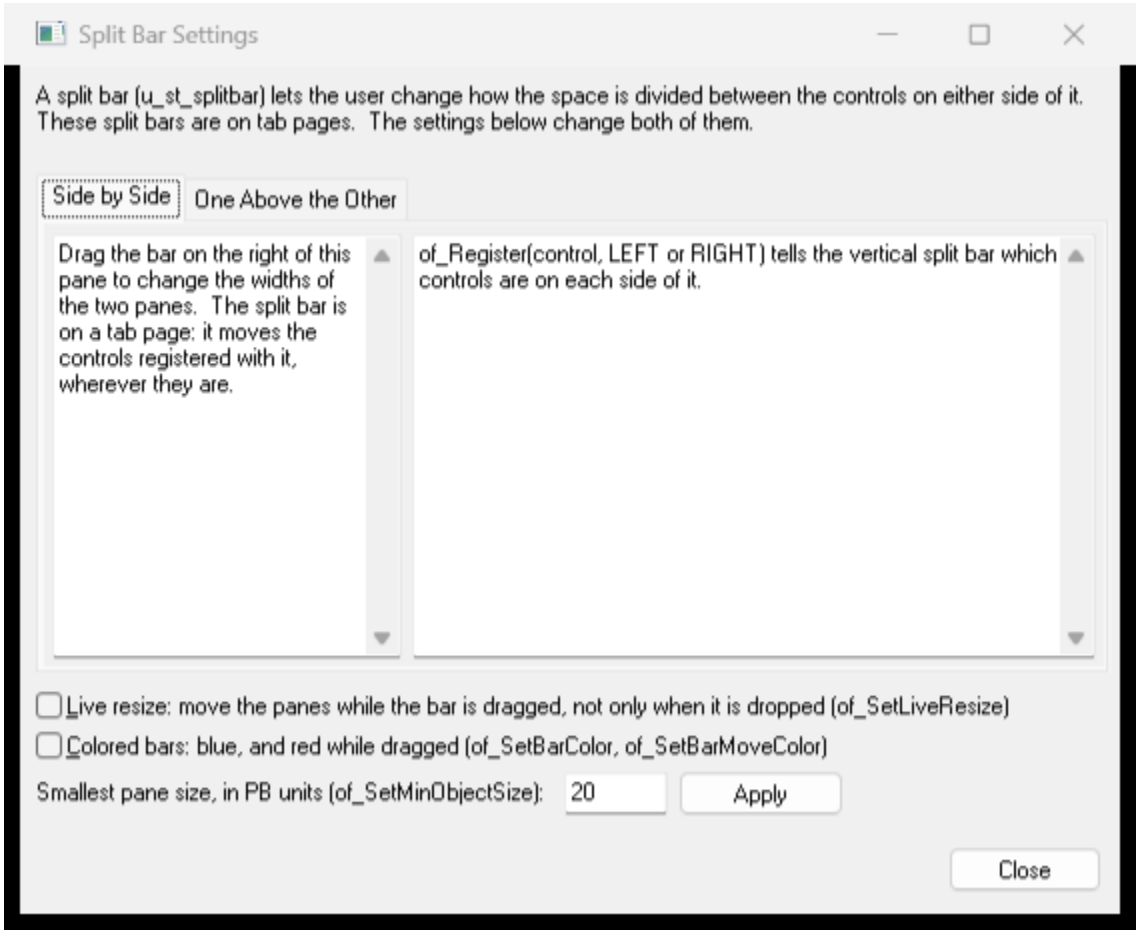
u_st_splitbar

The window

`w_splitbarexplorerstyle`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Split Bar Settings

Split bars on tab pages, and their settings: live resize (`of_SetLiveResize`), the bars' colors (`of_SetBarColor`, `of_SetBarMoveColor`) and the smallest size of a pane (`of_SetMinObjectSize`).



Try it

- Drag the bars on both tab pages, then change the settings and drag them again.

PFC objects

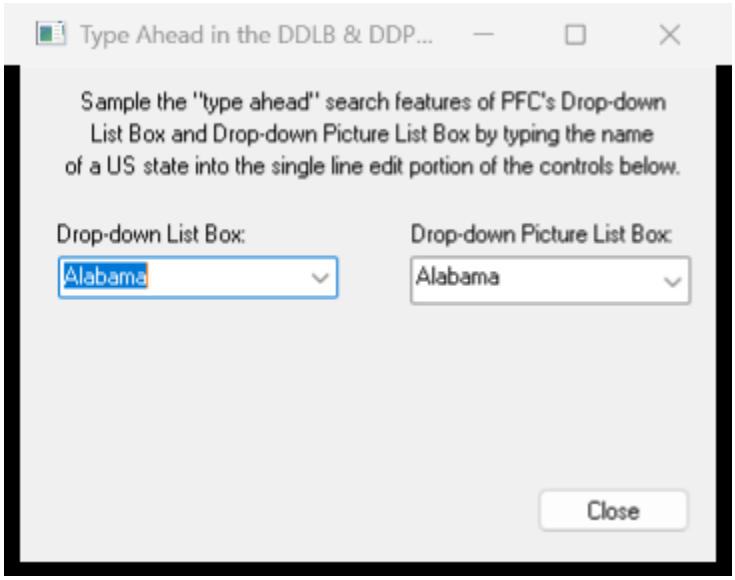
`u_st_splitbar`

The window

`w_splitbarwin`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.

Type Ahead in the DDLB & DDPLB

Type-ahead in PFC's drop-down list boxes (`u_ddlb`, `u_ddplb`): what the user types is completed with the first matching item.



Try it

- Type the first letters of a US state.

PFC objects

`u_ddlb`, `u_ddplb`

The window

`w_dropdownsearchlb`, in `appexamp.pbl`. Open it in the PowerBuilder IDE to see how the example is coded.